Domination And Exploitation: Understanding Industry Costs For Chicken, Egg, And Fish Products In The United States, Brazil, And China

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Background

As animal advocates know, mass torture and death are hallmarks of the animal agriculture industry. Such immense suffering would not exist without a market for animal products, requiring both buyers and sellers. A growing body of research explores the demand side of this market, with factors like price, taste, health, convenience, and social norms driving consumers' purchases (see Faunalytics, 2021; Rethink Priorities, 2023; Good Food Institute, 2023). However, far less research has examined the supply side of the market. In order for customers to have animal products to purchase, there must be companies creating those products in the first place.

The simplest way of understanding the horrors carried out in animal agriculture is through a consideration of the industry's profit motive. Like all businesses operating under global capitalism, the goal of the companies involved in the chicken, egg, and fish industries is to make money. By investigating the factors that shape profit-seeking actions by these industries, particularly the factors affecting their production costs, advocates can gain crucial knowledge about the industry they seek to change.

Industry Overview

Large animal agriculture corporations set their prices based on a variety of factors, the most important of which are supply and demand. The supply side consists of companies creating products while the demand side is made up of consumers who might buy those products. These two sides interact to create a market price that satisfies both groups.

Companies must also be able to cover the costs needed to make their products, otherwise they will not earn enough money to stay in business. So it's also possible to think of the price of a product as the sum of all of the costs of making that product plus some amount of profit for the company. Provided a company has a market for the product at that price, it can then determine how much consumers are willing to pay, which can affect the exact profit margin they receive. This can be seen in the recent rise in the price of eggs, which is due to various factors like increased feed costs, but also the continued demand for eggs even as prices rise.

Through this simplified lens, it can be easier to see the incentive companies have to reduce their costs. For example, if it costs a company $1.07 to produce a dozen eggs and grocery stores are willing to purchase eggs from producers at $1.25 per dozen, the egg company makes $0.18 per carton sold (Cal-Maine Foods Inc, 2023; USDA, 2023). If the company is able to lower its costs such that it only costs $1.00 to produce a dozen eggs, it will make $0.25 per carton sold. For companies selling over a billion cartons of eggs each year, like industry leader Cal-Maine, this small difference would add up to a staggering increase in profits.

Animal agriculture is a fairly low-margin industry, meaning companies make only a small bit of profit per dollar compared to many other large companies (Pi et al., 2014; Damodaran, 2023). For instance, experts often look at how much profit a company has left over after paying for

things like wages and materials (i.e., operating margin). Using this measure, the 500 largest companies in the U.S. averaged 17 cents in profit for every dollar they earned in 2022. By comparison, animal agriculture giant Tyson only kept 6 cents in profit per dollar earned (CSIMarket, 2023; Murphy, 2022; Tyson Foods, Inc., 2022). The remaining 94 cents of each dollar Tyson received in sales went to covering costs like labor and animal feed. Because its margins are already low, the industry is further incentivized to cut costs as much as possible in order to maximize profits. This desire to keep costs down is easily visible in animal agriculture. In most cases, the horrific conditions experienced by animals in industrial agriculture exist because companies have cut everything they can to increase their profit margin. Put another way, companies don't want to increase chickens' quality of life because it would cost too much.

Scope of Report

This report details many aspects that affect the cost of production in the broiler chicken, egg, and aquaculture industries in order to provide critical context for animal advocates. We focus on the ins and outs of these industries in three of the countries most central to animal agriculture: the U.S., Brazil, and China.

We hope that this report will provide a useful foundation for advocates to identify industry pressure points that could ultimately be used to increase production costs and improve animal welfare. Although outside the scope of this report, pairing these strategies with ongoing efforts to promote plant-based alternatives could lead to a decline in the power of the animal agriculture industry and the creation of a global food system no longer based on animal suffering.

Because of the global nature of animal agriculture, much of the basic economic structure of the industry is consistent across the countries covered in this report. As a result, this analysis will begin with a discussion of the broader industry on a global scale, then by considering the U.S. industries, for which most information is publicly available. We then turn to Brazil and China, whose industries resemble those of the U.S., but contain several key differences.

Terminology & Topic Coverage

The following report adopts industry and economic terminology, including phrases that obscure the realities of widespread animal suffering (e.g., "processing" animals). We chose this approach for two primary reasons. First, these terms are often more succinct ways of describing complex concepts or processes, especially where no commonly-used alternative terms exist. Second, this may help advocates familiarize themselves with the language used by producers, processors, government officials, and other industry players, making them more formidable opponents.

This report discusses economic concepts at a level intended for people who are new to them. Thus, experts may note that rather than a comprehensive exploration of terms, we provide simplified overviews to highlight the most relevant aspects and to provide direction to advocates

who may be interested in exploring certain topics further. These terms are used more frequently in the beginning of the report where we provide an overview of the global animal agriculture industry.

In addition, this report does not go into significant detail about the countless welfare issues that exist throughout the animal agriculture industry. Although some are discussed, most are outside the scope of this report and have already been excellently detailed in other publications. Instead, we widen our view to touch on as many aspects of the industry as possible in order to highlight more potential points of intervention for animal advocates.

Key Findings

1. **The fight against animal agriculture is an antitrust fight.** Competition is low in the global animal agriculture industry, with a small number of giant companies that control most of it. Corporations like Tyson or JBS are highly "vertically integrated": in addition to slaughtering billions of animals a year, they often own the mills that grind corn into feed, the genetics companies that breed fast-growing chickens, hatcheries where chicks are born, flocks comprising millions of birds, slaughterhouses, processing plants, and trucks that deliver products to stores. Integrating all of these stages under one giant corporation makes the production process cheaper and enables these companies to create staggering amounts of animal suffering. These megacorporations have also consolidated their markets by absorbing competitors or driving them out of business, producing a consumer landscape in which seemingly different brands—Jimmy Dean, Hillshire Farm, Ball Park, Sara Lee, and more—are actually all Tyson products. Other companies simply can't compete, leaving global giants like Tyson and JBS to dictate the rules and giving farmers little choice over which companies they work with.

2. **The U.S., China, and Brazil are key to the chicken, fish, and egg industries.** Not only are they global leaders in production, imports, and exports of animal products, they also literally feed the animal agriculture industry. As one example, Brazilian soybeans are used to feed Chinese fish that are ultimately eaten by U.S. consumers. An understanding of how the animal agriculture industry functions requires advocates to be familiar with the role these three countries play.

3. **The aquaculture industry hasn't yet consolidated or standardized as much as the broiler chicken and egg industries have, but it will.** Intensive aquaculture is relatively new and uses a wider variety of animals and production methods so it hasn't yet achieved the same level of efficiency as some other animal agriculture industries. However, this advancement will almost certainly happen in the coming years if nothing is done to stop it. With this intensification comes bigger, more concentrated operations, leading to even more animal suffering. Around the world, many governments consider aquaculture the future and are creating programs and initiatives to support its continued

rise. Without intervention, aquaculture companies will consolidate, vertically integrate, and intensify their operations.

4. **Governments have not only allowed but also encouraged animal agriculture to grow to this point.** In the U.S., companies have benefited from indirect subsidies that lower the cost of animal feed, as well as a regulatory environment that has allowed megacorporations to emerge and have considerable influence over the policies that are supposed to regulate them. In Brazil and China, the governments have directly supported animal agriculture with money and programs to promote the growth of companies they favor.

5. **Animal agriculture corporations' profits are sensitive to many risks.** Despite their power, these corporations are subject to a number of risks. In the U.S., current regulations require the ones that are publicly traded to disclose these risks as potential threats to their business so we have some insight into them. Those potential threats include consumer demands for better animal welfare, strengthened environmental policies, having to increase employee wages, and the loss of companies that are major customers.

6. **The modern model of animal agriculture even hurts the farmers that work for it.** Contract “grow-out” farmers (who raise the chickens that the megacorporations own) must often take out massive loans to meet those companies’ increasing demands. Even if they wanted to change industries, these farmers get trapped in debt and have little choice but to keep working for that industry. In the U.S. and Brazilian broiler chicken industries, a “tournament system” also pits these farmers against each other and lowers the pay for many. Some experts also fear that the rise of aquaculture could lead to further international exploitation of farmers.

7. **Animal feed is the biggest cost the animal agriculture industry has to cover.** While this always tends to be true, it has become more extreme over time: Feed now often makes up two-thirds of the money corporations spend to make animal products. Feeding the animals is a necessity, but because it is so costly for them, companies are looking for ways to make animals grow larger or produce more eggs with smaller amounts of food. They also benefit from government policies that make feed ingredients cheap. However, just because feed is the industry’s biggest expense doesn’t mean it’s the only thing advocates should focus on. These companies constantly look for cost-saving measures in every aspect of their business, from animal housing to labor costs to slaughter methods. Welfare-focused reforms like cage-free and slower slaughter line speeds aim to help animals but also reduce the industry’s profits. Higher wages for employees and tougher environmental regulations can do the same.

Recommendations

For advocates:

- **Learn the basics of how animal agriculture makes its money.** The complexity of economic systems can make this intimidating, but the colossal amount of suffering animals experience every day comes as a direct result of the industry's pursuit of profit. A basic familiarity with the factors that affect the cost of production for various animal products gives animal advocates a map of the industry. Through a careful examination of this map, we can identify points of intervention.

- **Act now to prevent aquaculture from becoming as concentrated and abusive as other animal agriculture industries.** The chicken and egg industries, for example, have already become highly efficient, industrialized, and standardized animal exploitation machines. Because aquaculture companies will try to model themselves after the companies that already dominate other industries, advocates can predict the paths they will take. For instance, one of the main ways this could happen is if seafood processing companies acquire their feed suppliers. This knowledge provides a rare opportunity to take action to prevent expansion rather than respond to an already entrenched industry.

- **Think of aquaculture as a collection of subindustries rather than a single industry.** There are major differences between aquaculture methods across species. What is important to salmon production, like moving fishes from freshwater to cages in the ocean as they mature, may not have any relevance in the tilapia industry. These subindustries work with very different animals and must use different practices as a result. In order to help the fishes raised in these systems, advocates need to learn about the ways particular species are raised.

- **Use what we know about supply and demand in the global economy to create change, and think beyond animal products.** Because the animal agriculture industry is globalized, customers in one country can make demands of companies in another. Advocates living in major producer countries like the U.S., Brazil, and China, or those living in regions with large amounts of purchasing power, like the E.U., can and should seek to change industry practices around the globe. One approach leverages consumer preferences in a country like Germany to demand higher welfare standards for imported animal products from other countries. Another is to focus on the cost and availability of feed—typically corn and soybeans—which is also highly globalized, perhaps finding other markets for those products or working to influence import tariffs.

- **Demand stronger, actionable antitrust regulations from governments.** Industry concentration has allowed a small number of megacorporations to dominate animal agriculture. Antitrust laws exist to encourage competition, protect consumer welfare, and

keep prices low. Demanding tougher regulations that limit giant corporations' continued growth, or even require existing megacorporations to be broken up, could weaken their stranglehold on the economy, including the global market for animal products.

- **Work to increase subsidies for plant-based products and their supply chains.** Governments around the world directly and indirectly support animal agriculture corporations in their countries with subsidies, cheap loans, and more. Similar support for plant-based agriculture that is intended for human, not animal, consumption would make produce and plant-based products more affordable as well.

- **Work with regional governments or labor groups to be more impactful.** Advocates working to change particular industries may have more success if they target the areas where those industries have a strong presence instead of only working at the federal level. For example, the Brazilian broiler industry is concentrated in the South region, but the egg industry is more active in the Southeast. In the U.S., the South is home to both the broiler industry, as well as the country's young aquaculture industry. In China, the broiler industry is growing in Shandong Province, but egg production is spread across the North and Northeast, while aquaculture practices vary in different parts of the country. Rather than taking a one-size-fits-all approach, advocates could be more effective by focusing on areas where each industry has the largest presence.

For researchers & funders:

- **Fund, conduct, and translate research on the animal agriculture industry.** There is a lack of high-quality information on the ins and outs of the animal agriculture industry that is written by or for animal advocates. Much of the existing research focuses on vertical integration, consolidation, and feed, is available in a single language (often but not always English) and is either academic or written by industry insiders, making it less useful to advocates. Advocates need more research about the specific processes and costs involved in the production of animal products, for existing research to be made available in languages including Portuguese, Chinese, and English, and for advocates with knowledge of the industry to create short, simple resources describing industry pressure points. These recommendations are particularly pertinent to aquaculture, which employs many different methods to exploit a wide range of animals, making it more difficult to develop high-impact strategies.

- **Collaborate with other groups that have shared goals.** Organizations focused on climate change and the environment often have similar goals as animal advocates. For example, both groups would like to see the industry pay to limit and address the effects of its environmental destruction. Increased government oversight would also increase animal agriculture companies' costs. Labor groups and exploited contract farmers are more potential allies, as improving worker pay, benefits, and conditions through unionization and regulation would also increase costs for the animal agriculture industry.

- **Research ways that the importance of feed can be leveraged to help animals.** Animal agriculture companies spend most of their money on feed, so any policy change that affects the feed industry will also affect the chicken, fish, and egg industries. How can advocates use this fact to change industry practices? Policies related to fertilizers, monocultures, and the environmental effects of the grain industry may be places to start. The dominance of companies like Bayer, DuPont, and other giants in the genetically engineered seed industry could also be worth exploring.

- **Explore how price affects purchases for conventional, cage-free, and free range eggs.** Demand for eggs is relatively "inelastic": Increases in their price don't greatly affect the number of eggs consumers buy. However, there is limited research on how that applies across different types of "speciality" eggs so we don't know how egg consumption will be affected by new welfare practices increasing prices. Research on the cross-price elasticity of different egg types could help advocates determine which strategies might be most effective at decreasing animal suffering.

- **Analyze how future industry changes could affect animals.** Each animal agriculture subindustry researches ways to cut their costs, especially with new technologies or practices. The animal advocacy movement can conduct its own research to anticipate how these innovations will affect chickens, fishes, and other animals. This knowledge could ultimately help advocates prevent harmful practices from being implemented.

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Terms To Know

Oligopoly

A market in which a small number of producers are dominant.

In animal agriculture: The U.S. and Brazilian chicken industries resemble oligopolies. Only a few titans, like JBS and Tyson, control a huge proportion of the market.

Vertical Integration

A business strategy in which a company controls various stages of the supply chain.

In animal agriculture: Many animal agriculture corporations are vertically integrated, owning the feed mills, slaughterhouses, processing plants, and transport trucks.

Antitrust Laws

Regulations that encourage competition by limiting the market power of any particular company.

In animal agriculture: Current antitrust laws have done little to prevent a few massive corporations from dominating the chicken and egg industries.

Price Elasticity Of Demand

A measure of how much more or less of a product people buy if the price changes.

In animal agriculture: In the U.S., demand for eggs is less elastic than demand for fish or chicken. In other words, people are more sensitive to price changes for fish and chicken than they are for eggs.

Integrators

In animal agriculture: Large broiler chicken companies are known as integrators. They outsource flocks of chickens to various farmers who raise the chickens.

The integrators then combine or integrate the flocks as part of the slaughter and processing stages of meat production.

Economies Of Scale

Cost savings that occur when companies increase production or become more efficient.

In animal agriculture: This motivates companies to become larger, which causes increased animal suffering.

Applying These Findings

We understand that reports like this have a lot of information to consider and that acting on research can be challenging. Faunalytics is happy to offer pro bono support to advocates and nonprofit organizations who would like guidance applying these findings to their own work. Please visit our virtual Office Hours or contact us for support.

Behind The Project

Research Team

The project's lead author was Zach Wulderk. Dr. Jo Anderson reviewed and oversaw the work.

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Research Terminology

At Faunalytics, we strive to make research accessible to everyone. We avoid jargon and technical terminology as much as possible in our reports. If you do encounter an unfamiliar term or phrase, check out the Faunalytics Glossary for user-friendly definitions and examples.

Research Ethics Statement

As with all of Faunalytics' original research, this study was conducted according to the standards outlined in our Research Ethics and Data Handling Policy.

Method

We carried out this exploratory study with a thorough investigation of economic, political, and industry factors affecting broiler chicken, layer hen, and aquaculture production around the world, focusing on some of the countries with the highest production totals.

The countries discussed in this report—the United States, Brazil, and China—are consistently among the world's largest contributors to the global slaughter of chickens, fishes, or both. Each is also a major player in the trade of animal feed and the international market for chicken, egg, and fish products (FAO, 2023a). Several of the largest animal agriculture companies in the world are also based in these countries, including Tyson and Cal-Maine in the U.S., JBS and BRF in Brazil, and WH Group, Wen's Food Group, and Wellhope Agri-Tech in China.

In other words, these three countries are undeniable forces in the global exploitation of animals. By reviewing industry, economic, and academic analyses and data related to the target industries and countries, we are able to provide summaries of many of the primary factors affecting the cost of production of chicken, egg, and fish products.

An Overview Of The Global Animal Agriculture Industry

The U.S. slaughters over 9 billion chickens each year. Unfathomably large numbers like this rightly prompt animal advocates to take action, targeting U.S. corporations like Tyson and seeking to shift the preferences and purchasing habits of U.S. consumers. But what sometimes gets lost or overlooked in these efforts is the global nature of the animal agriculture industry. More than 16% of the poultry produced in the U.S. is shipped overseas, meaning that a significant portion of the country's broiler industry isn't focused on domestic (U.S.-based) consumers. Similarly, in Brazil, the world's largest exporter of poultry, almost 30% of its poultry products are sent to other countries (FAO, 2023a). And well over half of global aquaculture production takes place in China, the world's primary seafood exporter (Crona et al., 2020; FAO, 2023a).

This report focuses on several of the biggest producers of animal products, but it is important to note that producers can be in one country while consumers are in another. For instance, Japan, Mexico, Germany, and the United Kingdom are all major importers of chicken. Several Northern European countries, including Germany, the Netherlands, Belgium, and France, are among the top egg importers. Japan and Germany are also among the biggest fish importers, as is Thailand (FAO, 2023a; OEC, 2022a; Ngamprasertkit, 2018). Customers for animal products in these countries are an integral part of the continued growth of animal agriculture in places like the U.S., Brazil, and China, and suppliers around the world alter their production practices to meet the demands of consumers.

No matter which area of animal agriculture you explore, you'll run into the U.S., China, and Brazil. For example, in 2020, these three countries were the world's biggest producers of poultry meat. They are also frequently among the most common producers, exporters, and importers of other animal products, including eggs and fishes. The U.S., Brazil, and China are also home to a small number of animal agriculture megacorporations that collectively dominate the global animal product trade. The economic and political environments of these countries often support the industry's growth.

Globalized trade isn't limited to chickens. Discussed in more detail below, the most expensive aspect of most animal agriculture industries is feeding the animals, and one of the primary ingredients involved in this is the soybean. Brazil is both the world's largest producer and largest exporter of soybeans. In fact, more than two-thirds of the soybeans grown in Brazil are sold abroad (FAO, 2023a). The majority of these end up in China, where one of the main ways these soybeans are used in China is to feed aquacultured fishes (Godfrey, 2021). China is a global leader in fish exports and the U.S. is one of the major importers (FAO, 2023a). In other words, Brazilian soybeans are used to feed Chinese fish that are ultimately eaten by U.S. consumers.

A Few Megacorporations Control The Global Market

Advocates are not wrong to say that the U.S. slaughters the most chickens in the world, but this point misses important context. It is more accurate to say that Tyson and Pilgrim's Pride slaughter the most chickens—but one of those corporations, Pilgrim's Pride, is owned by a Brazilian company. We highlight this not to shift blame from one country to another, but to demonstrate how the animal agriculture industry is dominated by a few massive corporations operating in many different countries around the world. As a result, animal advocates need to choose targets and strategies with large corporations and the global trade of animal products in mind.

The second-largest poultry processor in the U.S. is Pilgrim's Pride, which "processes" (kills) nearly 8 million chickens every day (Pilgrim's Pride Corporation, 2022). Although Pilgrim's Pride is a U.S.-based company, more than 80% of it is owned by the Brazilian megacorporation JBS (Pilgrim's Pride Corporation, 2022). In fact, JBS is the largest meat corporation in the world and one of the world's largest broiler chicken producers (JBS, n.d.; Clements, 2023). Considering all the companies it owns (i.e., subsidiaries), JBS is also the largest beef and second largest pork producer in the U.S. (JBS Foods, 2023). In addition to its majority ownership of Pilgrim's Pride and other subsidiaries in the U.S., JBS and its subsidiaries also operate in Canada, Mexico, the United Kingdom, France, the Netherlands, Australia, and New Zealand (JBS Foods, 2023).

The worldwide footprint of JBS underscores the globalized nature of the meat industry. Similarly, the U.S.-based company Tyson has facilities in 16 countries and China-based WH Group owns companies like Smithfield, the biggest pork company in the U.S. (Tyson Foods, Inc., 2021; Sentient Media, 2021).

Alongside legislative campaigns and other efforts with a domestic focus, it is important for advocates to be aware of the global economy in which companies like JBS operate because while it adds complexity to advocacy, it also adds opportunities. Every year, for example, decisions by the Brazilian government influence the slaughter of billions of chickens in the U.S., while the demands of U.S. consumers support fish production in China. Although it may have hubs in certain countries, the animal agriculture industry comprises a network of corporations spread across the globe. Because of this, domestic efforts can have international implications, a reality that animal advocates can leverage to maximize their impact.

Feeding Animal Agriculture

Animal feed is the central piece of the global animal agriculture industry. As meat consumption continues to increase in much of the world, it is critical for advocates to understand the role that the feed crops, particularly corn (also known as maize) and soybeans (also known as soya beans), play in fueling this growth (Weis, 2013; Sharma, 2014). Corn and soybeans are absolutely central to the animal agriculture industry. About 37% of cereal crops grown around the world are fed to confined animals (Our World In Data, n.d.). More specifically, 77% of soybeans and 56% corn worldwide are used for animal feed (Ritchie & Roser, 2021; Erenstein et al., 2022). The grain and animal agriculture industries are highly interdependent, with the growth of one fueling the growth of the other. Policies that support the expansion of the feed grain industry also support the expansion of the animal agriculture industry because abundant feed allows animal products to be made more cheaply (Winders & Ransom, 2019; Sharma, 2014).

The cost of feed is usually the single largest contributor to the total cost of an animal product. For that reason, the industry spends significant resources researching ways to make feed cheaper and to make animals grow larger or produce more with less feed.

Although feed is always a major cost for producers, when feed is the *primary* cost in production it can often indicate that an industry has grown highly efficient with regard to other costs such as labor and machinery—that it is a highly advanced and industrialized industry (Asche et al., 2018). Put differently, feed is the biggest cost in these industries because they've been able to minimize the cost of nearly everything else involved in their production processes. For example, feed costs in the Norwegian salmon industry grew from about a quarter of total costs to half of total costs between 1986 and 1998 as the industry grew more advanced. This occurred even as the price of feed plummeted because technological innovations helped reduce other costs (Guttormsen, 2002). However, the importance of feed doesn't mean that it's the only part of production advocates should focus on. The industry needs many other costs—including labor, energy, and compliance with animal welfare regulations—to be cheap in order to maximize its profits.

The U.S., Brazil, and China are the main players in the international trade of feed grains. Even if they didn't also lead the world in direct animal exploitation, their collective role in feeding the industry would make them key accomplices to animal agriculture. Two of the most important

ingredients in farmed animal feed are corn and soybeans. In 2020, the U.S. and Brazil were the top producers and exporters of both corn and soybeans. China was also a global leader in corn and soybean production as well as one of the biggest importers of both crops (FAO, 2023a). In other words, these three countries are responsible for feeding the animal agriculture industry.

Because of how globalized the agriculture industry has become, the quality of a feed crop in one country can affect the costs of broiler chicken, fish, or egg production in another country. The price of corn and soybean ingredients for animal feed are shaped by factors from the weather to government policies around ethanol production (Weaver, 2014).

Just as the production of primary feed ingredients is dominated by a handful of countries, it's also dominated by a handful of multinational corporations, paralleling the animal agriculture industry itself. These megacorporations include four that are often lumped together as the "ABCD" companies (ADM, Burge, Cargill, and Louis Dreyfus) and COFCO, a state-owned corporation that is China's largest importer and exporter of corn, soybeans, and many other crops (COFCO Corporation, 2017; Favas, 2019). In 2016, the ABCD companies alone controlled about 80% of the global soybean trade (Schneider, 2019). Some of their biggest customers are animal agriculture companies, so these massive grain trading companies also have a vested interest in using their power and influence to lobby for policies that benefit meat producers. Because the animal agriculture industry spends so much of their money on feed, the factors that affect the grain giants are also factors that affect the animal agriculture giants. Cargill may be the worst example: it is both a leading grain trader and a leading animal agriculture company, underscoring how interconnected these two industries are (Reuters, 2021).

In this report, we don't discuss most of the key factors in the cost of crops like corn and soybeans in order to maintain our focus on animal agriculture. However, because of the outsized influence of feed inputs on the cost of chicken, egg, and fish production, further research should explore topics such as seed genetics, fertilizers, monocultures, and the general environmental effects of the grain industry. All of these could provide important insights for animal advocates, as well as opportunities for cooperation with other advocacy movements, such as the climate movement.

Feed Conversion Ratios

In order for advocates to understand how feed shapes the industry, we need to understand how producers think about it. Because feed tends to be their primary cost, producers of animal products want to make it go as far as possible. In this context, the animal agriculture industry will often discuss measures of how efficient animals are at "converting" feed into the desired product—typically, their own flesh. For example, the broiler chicken industry is concerned with how much feed is needed for a chicken to produce a certain amount of meat, a measure known as a *feed conversion ratio*.

In the past, chickens were selectively bred to grow large. More recently, breeders have begun selecting chickens that have the most efficient feed conversion ratios (Prakash et al., 2020;

Asche et al., 2018): that is, the ones who need less food to grow. This helps the industry reduce their main cost. The chickens they breed will convert as much of the feed they consume into meat as possible, as opposed excreting it as waste or using it for other basic bodily functions. Producers also work to identify ideal feed blends and carefully manage environmental factors to ensure that the chickens' bodies maximize meat production.

As animal agriculture industries grow more advanced, they have more and more control over genetics, feeding, and environmental management, enabling them to improve their feed conversion ratios. For example, compared to the industry in the 1950s, the U.S. broiler chicken of today needs only about half of the feed to produce the same amount of meat (National Chicken Council, 2022). Similarly, salmon of the 1980s needed almost three times as much feed to produce the same amount of meat as salmon of the present (Asche et al., 2018).

This is important for understanding where animal agriculture companies are focusing their cost-cutting efforts. Because feed is usually their costliest expense, more efficient feed conversion ratios are among the industry's most consistent goals. Cheaper production allows the companies to grow larger and exploit animals in even greater numbers.

Tariffs, Duties, and Trade Deals

Because access to feed is so central to the animal agriculture industry, anything that can significantly impact the price of feed is important for animal advocates to understand. For instance, tariffs and duties are essentially government-imposed taxes on products imported from other countries, including animal products and feed. For example, the Chinese government may impose tariffs on corn imported from the U.S., making it more expensive in China and thereby encouraging Chinese buyers to purchase cheaper, domestically grown corn. This in turn reduces U.S. corn farmers' sales. Tariffs and duties don't only apply to feed crops and are important considerations for many goods, including animal products themselves.

Related to tariffs and duties are more comprehensive trade deals. One such example is the United States-Mexico-Canada Agreement (USMCA), the successor to the North American Free Trade Agreement (NAFTA). The USMCA outlines many aspects of trade between the three largest countries in North America, including for animal products. For example, the U.S. supplies about 85% of the chicken imported into Canada. Under the USMCA, the U.S. is allowed to export thousands of metric tons of chicken products into Canada with little or no tariffs or duties applied. Because of this, Canadian consumers purchase U.S. chicken products more often than similar products imported from other countries because they're generally cheaper. In other words, the USMCA essentially allows U.S. chicken producers to get more customers (USDA, 2022). More customers means more demand for chicken products, which ultimately increases chicken suffering. And it's not just meat: the USMCA also allows more egg exports from the U.S. to Canada.

Deals like the USMCA support global animal agriculture in other ways as well. For example, the overwhelming majority of yellow corn used in Mexico—mostly for animal feed—comes from the

U.S. without any tariffs under the USCMA (USDA, 2020). In other words, this deal also helps cut costs for Mexico's animal agriculture industry.

Trade agreements such as the USMCA (and NAFTA before it) are easily overlooked by animal advocates as ways of shaping the animal agriculture industry. They have a significant impact on trade between countries, which can affect demand and prices for both animal products and feed ingredients.

Price Elasticity Of Chicken Meat, Eggs, And Fishes

As advocates continue to pressure companies to change their policies, production costs will increase and prices tend to be hiked up as well to compensate. But how much do price increases affect the amount of animal products people purchase? We can try to answer this question by considering each product's *price elasticity of demand*, which is a measure of how much more or less of a product consumers buy when the price changes. For example, if the price of cheese goes up a few cents, how much less cheese do people buy? A variety of factors can influence how elastic (affected by changes in price) demand for a product is: whether other, similar products are available, the purchaser's income and preferences, and how essential that product is to people. When a price increase results in a big drop in purchases, demand for that product is considered to be elastic.

For example, consider two neighboring gas stations that offer fuel at the same price. If one gas station decided to increase its price by 10 cents per gallon, many customers would choose to buy gas from the cheaper gas station. In this case, the availability of a perfect substitute results in demand being *elastic*, or very easily affected by price changes. By contrast, most people with diabetes would continue to purchase insulin regardless of how much the price increases because it is a necessity. As a result, demand for insulin is highly *inelastic*.

Understanding the price elasticity of animal products can allow advocates to predict how much a change in the price of a product will affect the quantity purchased and ultimately the amount of animal suffering involved in its production. In the U.S., eggs are one of the most inelastic food products on the market: changes in price don't affect changes in purchase very much. Historically, a 10% increase in egg prices would cause less than a 3% decrease in the number of eggs purchased on average. Fish products have been more elastic, with a 10% price increase resulting in a 5% drop in purchases.

Of the three products covered in this report—chicken, eggs, and fish—demand for chicken and other poultry products is the most sensitive to price changes. Historically, a 10% increase in the price of poultry resulted in a nearly 7% drop in purchases. All three of these products are less sensitive to price changes than beef and pork products (Andreyeva et. al. 2010). Because demand for eggs is much less elastic than demand for poultry, egg prices would have to increase by much more than chicken prices to create a similar drop in quantity purchased.

Elasticity of demand differs from country to country and even region to region, so it is important for advocates to study how it may be shaped by different factors in their particular area. For example, in China, researchers estimate that a 10% increase in the price of eggs may result in a 6% decrease in the number of eggs purchased, more than twice the drop expected in the U.S. (Chen et al., 2015).

The U.S. Broiler Industry

The U.S. pioneered the broiler industry and in doing so ushered in the modern age of industrial animal agriculture. Today, the U.S. slaughters over 25 million chickens every day, more than any other country in the world with no signs of slowing down (Faunalytics, 2022). The average U.S. resident consumes 115 pounds (52 kg) of chicken and other poultry products each year, 15 pounds more than they did only 20 years ago (USDA ERS, 2023). The U.S. is also the second biggest exporter of chicken products in the world.

How A Few Corporations Came To Dominate The Broiler Industry

As discussed previously, a handful of megacorporations dominate the global animal agriculture industry. Some familiarity with the origins of this situation is important for understanding the current landscape and the ways the broiler industry and animal agriculture more generally could further evolve.

Horizontal And Vertical Integration

To explore the growth of the broiler industry, let's consider one of the largest animal agriculture companies in the world: Tyson Foods. Founded in the 1930s, Tyson began with one man raising chickens in Arkansas and selling them in cities like St. Louis and Chicago (Leonard, 2014). Nearly a century later, Tyson is one of the top 100 companies on the New York Stock Exchange, with revenues exceeding those of other corporate giants like Nike and Coca-Cola.

Tyson is currently the second largest meat processor in the world, a status it and its industry-dominating peers achieved via the concentration of market power (Fortune, 2023). They and companies like them achieved this status through processes known as horizontal integration and vertical integration, as well as expansion into other industries.

Horizontal integration is essentially the acquisition of competitor companies, like when one company that processes chickens purchases or merges with another company that processes chickens. Swallowing up competitors like this was a major contributor to the growth of the big animal agriculture companies like Tyson.

However, *vertical integration* may be an even bigger factor. This is the process through which a company acquires companies it previously purchased from or sold to. No matter how many competitors a broiler chicken processing company takes over, it still needs to buy fully grown

chickens from a farm to process, and then sell their meat products to a distributor. That is, unless it is vertically integrated.

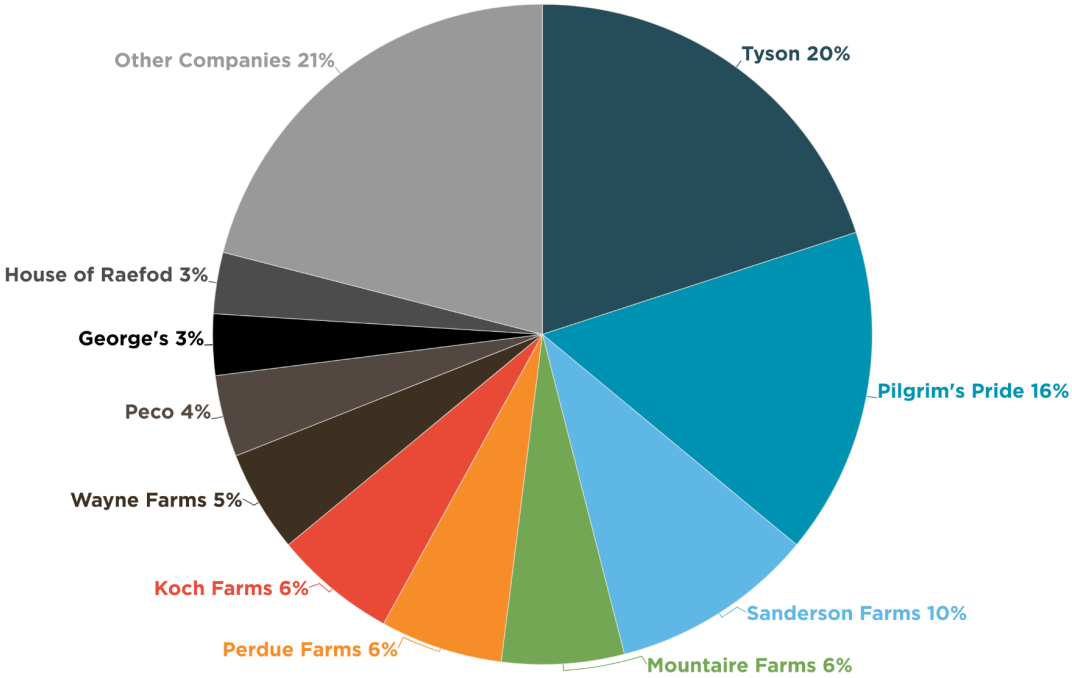
A "fully vertically integrated" company like Tyson owns not only the chickens, the processing plants, and the distribution network, but also virtually every other step in the chain. For example, Tyson owns the mills that process raw grain into feed for chickens. This feed is given to breeder chickens that come from a genetics company that is also owned by Tyson, and to chicks in Tyson-owned hatcheries that come from that same genetics company. When the chickens are large enough for slaughter, Tyson-owned trucks transport the chickens to Tyson-owned slaughterhouses before Tyson-owned facilities turn the slaughtered chickens into products like chicken nuggets that are then packaged and shipped by workers employed by Tyson (Tyson Foods, Inc., 2022). This affords them considerably more control, certainty, and efficiency in their business (Investopedia, 2021). Because vertically integrated companies don't have to worry about every company in the supply chain needing to make a profit, they can also save money at each stage of production.

Industry Concentration

Because vertically integrated companies have control over so many parts of the production process, they are able to exert a huge amount of power over the market. Vertical integration has made the U.S. broiler industry highly concentrated: there are only a few companies for customers to choose from.

The array of brand names in the grocery store hides the amount of control held by just a small number of companies. For example, Jimmy Dean, Hillshire Farm, Ball Park, and Sara Lee are just a few of the brands that are just Tyson subsidiaries (Tyson Foods, 2023). Tyson is the largest broiler company in the U.S. and produces over 20% of the country's broiler chicken products. The second and third largest, Pilgrim's Pride and Sanderson Farms, control about 16% and 10% of the market, respectively. This means that only three companies account for almost half of all of the chicken products sold in the U.S. (Bolotova, 2022).

Figure 1. Share Of U.S. Broiler Chicken Market By Company



Source: Bolotova, 2022

When a small number of companies dominate the market for a product, they are able to exert considerable control over the production process and prices. This type of market is considered an *oligopoly*. Like a monopoly but with a few more sellers, oligopolies allow companies to exert economic power in ways that they wouldn't be able to with less concentration. In an oligopoly, there are high barriers to entry, meaning that it would cost new companies exorbitant amounts of money to reach competitive status with the current giants of the industry. This includes things like building processing plants, setting up hatcheries, and acquiring access to feed.

In addition to making it difficult for other animal agriculture companies to challenge the current industry leaders' positions, this also poses difficulties for plant-based producers who currently have significantly less production capacity. For example, in all of 2022, about 255 million units of plant-based meats were sold (Good Food Institute, 2023). Tyson alone could process the same amount of chicken in under six weeks (Tyson Foods, Inc., 2022). Even if demand for plant-based meats rivaled demand for chicken, plant-based companies would need to invest huge amounts of money in scaling up their operations, developing efficient machinery, and finding workers in order to match the output of companies like Tyson.

Another downside of oligopolies is the potential for illegal price fixing, which typically occurs when companies secretly work together, agreeing to sell their products at a specific, higher price

than would naturally occur through the mechanisms of supply and demand. This practice is commonly outlawed because it artificially increases prices. In recent years, the broiler industry has faced numerous lawsuits related to price fixing and, as of early 2022, had paid out over $360 million in settlements (Bolotova, 2022).

Concerns about industry dominance are nothing new. As U.S. corporations like Standard Oil began to grow and monopolize markets for various products in the late 19th and early 20th centuries, the federal government passed a series of laws, known as antitrust laws, in order to promote competition and benefit consumers through lower prices and better quality products (Federal Trade Commission, n.d.). Over time, and especially as policymakers with strong beliefs in market deregulation and other neoliberal policy approaches came to power in the 1970s and 1980s, antitrust enforcement became less of a priority for the government (Vaheesan, 2017).

Undeterred by antitrust laws, the broiler industry has gotten more concentrated over time. In 1986, the top four companies controlled 35% of the poultry market (Kelloway and Miller, 2019). By 2020, that number had climbed to 53% (Bolotova, 2022). Experts will sometimes describe a market as an oligopoly if the five biggest companies account for over 60% of the sales in that market (Hennerich, 2023). And as of 2020, the U.S. broiler industry was extremely close to hitting this mark, with 59% of the market owned by Tyson, Pilgrim's Pride, Sanderson Farms, Mountaire Farms, and Perdue Foods (Bolotova, 2022). In other words, the modern U.S. broiler industry closely resembles an oligopoly.

Other issues can arise when there are only a few buyers for a product or service. For instance, a small group of buyers, like chicken companies, can make demands of producers, like contract chicken farmers. Producers must meet these demands, like very low prices, because they have few other buyers to turn to. This type of situation, where there are only a handful of powerful buyers, is essentially the flip side of an oligopoly.

"Grow-out" farmers—those who contract with the big chicken companies to raise chicks up to slaughter age—often don't have multiple companies to choose from when selling their services either. Instead, they must contract with large companies like Tyson, which are known as "integrators" because they combine and process the chickens each of their grow-out farmers raises (RAFI USA, 2023). Because it is difficult to transport live chickens great distances, these farmers must typically accept whatever price they're offered, and most have just one or two options (Hamilton and Sunding, 2020; MacDonald and Key, 2012). In other words, industry concentration hurts farmers as well because it allows the broiler companies to pay them almost as little as they want.

Influence On The Industry And Government

The handful of big companies that dominate the broiler industry have such a stranglehold over the market that they have the money and power to set low industry standards for animal welfare, worker treatment, pollution, and more, keeping costs down (Harvard Law Review, 2004). For instance, if one company increases the speed at which it slaughters chickens, all of

the other chicken processors must also speed up their slaughter process or risk falling behind the more efficient company (Harvard Law Review, 2004; Schlosser, 2001). Through actions like this, the whole industry quickly adopts the cheapest practices allowable by law, usually increasing both human and non-human animal suffering in the process.

These companies also shape the regulations that govern their practices. The money they have accumulated allows them to pay lobbyists to influence legislators and regulatory agencies into creating laws and policies in their favor (Boehm, 2021).

This influence on the government can also be seen in the "revolving door" of animal agriculture industry lobbyists-turned-regulators at the U.S. Department of Agriculture (USDA; Friedrich and Wilson, 2016). According to one investigation, "Big Agribusiness has been able to pack [the] USDA with appointees who have a background of working in the industry, lobbying for it, or performing research or other functions on its behalf" (Mattera, 2004). This essentially allows the broiler industry and other animal agriculture industries to regulate themselves.

Expansion Into Other Animal Agriculture Industries

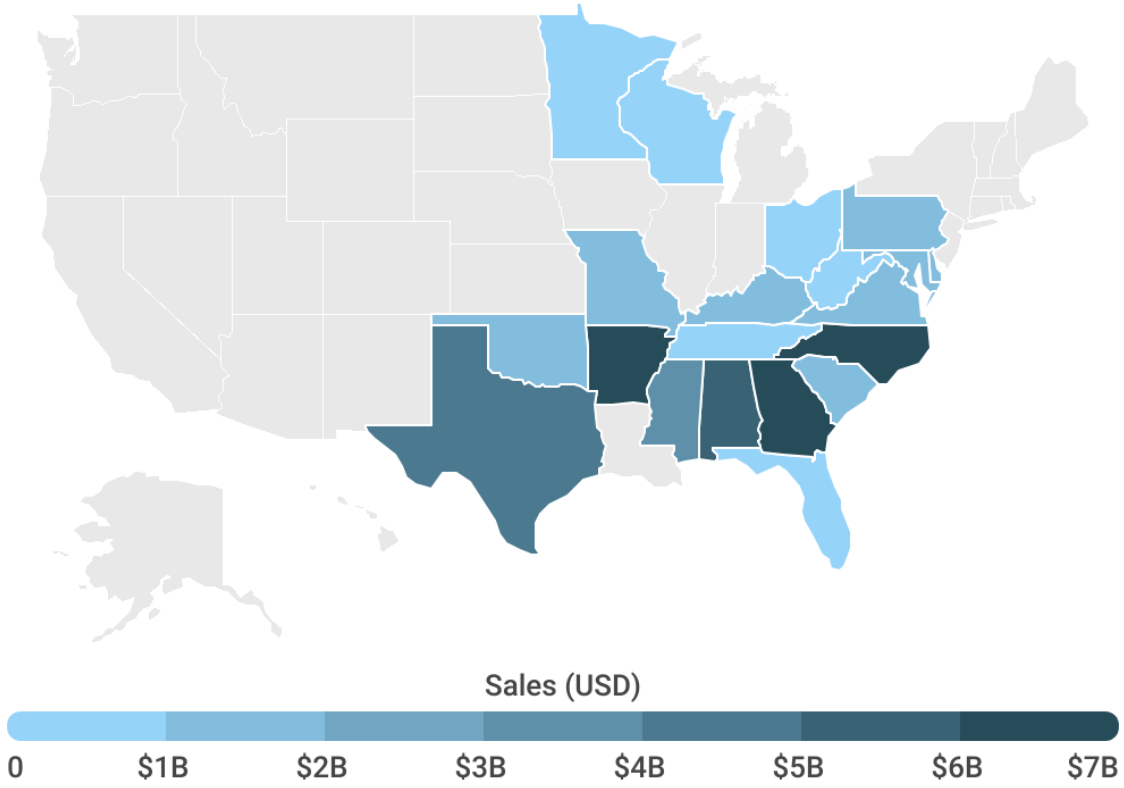
These companies have also branched out into processing other animals, further increasing their power and making them integral to the economies of not only the U.S., but also several other major economies around the world, like Brazil and China. For example, Tyson's beef products bring in more money than its chicken products and U.S.-based Pilgrim's Pride is primarily owned by Brazilian company JBS, the world's largest meat producer (Tyson Foods, 2021). As advocates have told Faunalytics in the past, challenging any aspect of the colossal meat processing companies' production process is often a difficult undertaking because of the amount of money and power they have at their disposal.

The Basics Of Broiler Chicken Production

In the U.S., poultry processing—which primarily involves chickens, but also turkeys and several other birds—is dominated by four firms (USDA APHIS, 2022; The White House, 2022). Together, Tyson, Perdue, Sanderson Farms, and Pilgrim's Pride control roughly 54% of the poultry products sold in the U.S., the vast majority of which come from broiler chickens (The White House, 2022; USDA ERS, 2023). The broiler industry produced nearly 21 billion kg of chicken products and earned them $50 billion in revenue in 2022 alone (USDA ERS, 2023). About 16% of this was exported abroad (USDA ERS, 2023).

The U.S. broiler chicken industry is concentrated in the South, primarily in North Carolina, Georgia, Arkansas, and Alabama. In 2022, they produced an average of 1.1 billion broiler chickens per state, yielding between $5.7 billion and $6.9 billion in sales each (USDA, 2023; USDA ERS, 2023).

Figure 2. Sales of Broiler Chickens Among Top Producing States, 2022



Source: USDA ERS, 2023
Notes: States colored blue gray for $3.2 billion in broiler chicken sales in 2022. 6% of national sales are not disclosed to avoid disclosing individual operations.

The Production Process

Production of broiler chickens begins with feed. We discussed the overall importance of feed in global agriculture above, but here we get into the specifics of feed within the U.S. broiler industry. By making these steps and specifics clear, we hope that advocates will be able to identify points where intervention to increase costs or improve animal welfare is possible.

Feed mills, which convert raw products from farmers into the ground-up meal that is used in feed, are typically owned by integrators (i.e., big broiler chicken companies; Weaver, 2014). Corn is the primary ingredient in broiler chicken feed, with estimates putting the proportion of corn at almost 45% (Unveren et al., 2020; Decision Innovation Solutions, 2017). Soybeans are the second most important ingredient, making up about 26% of broiler chicken meal (Unveren et al., 2020; Decision Innovation Solutions, 2017). The remaining components include bakery meal (14%)—essentially a ground-up mixture of extra human foods like cereal and cookies—and several other ingredients (Decision Innovation Solutions, 2017; Scoular, 2022). Together, the various ingredients used in feed account for between 65% and 75% of the total cost of

production of broiler chickens (Weaver, 2014; Tyson Foods, Inc., 2022). However, this cost can vary greatly based on feed crop harvests and markets.

Feed goes to two places in particular, the first of which is the breeder farm. Here, "grandparent" chickens are bred to produce fertilized eggs that will hatch into broiler chickens (Tyson Foods, Inc., 2022). Breeders use a process that selects for traits that will keep costs down (e.g., a highly efficient feed conversion ratio, as discussed in that section above) or yield a more valuable product (e.g., more breast meat; Weaver, 2014). A small number of animal genetics companies dominate the breeder industry. For instance, Aviagen controls around 45% of the global breeder market and Tyson-owned Cobb-Vantress also controls a huge portion of the market (USDA APHIS & Iowa State University, 2022; Della Rosa, 2016). Despite the different varieties of chickens these companies advertise, including the Cobb, Ross, and Arbor Acres "strains," these are all ultimately crosses between White Cornish and White Plymouth Rock chickens (The Livestock Conservancy, 2020; Laatsch, n.d.).

Integrators acquire eggs from "grandparent" chickens raised on breeder farms. These eggs are then moved to hatcheries, which are usually owned by the integrators, and where they are kept in special incubators so that the eggs hatch properly. Vaccination also occurs here shortly after the chicks hatch (Weaver, 2014).

Newly-hatched chicks, called day-old chickens (DOCs), are then transferred to grow-out farms, where they are raised until they are large enough to be sent back to the integrator for slaughter and processing, usually at less than 7 weeks old (National Chicken Council, 2022; Weaver, 2014). While some companies like Tyson own their DOCs from the beginning, other companies like Pilgrim's Pride purchase their DOCs from breeders (Tyson Foods, Inc., 2022; Pilgrim's Pride 2022 10-K; Polansek, 2021).

Grow-out farms are the other point in the broiler production process where feed is used. In addition to feed and eggs from breeders, other main costs for the broiler chicken industry include "water, housing, labor, medicine, and machinery," though specific numbers for these inputs are less readily available (Weaver, 2014, p. 17).

Slaughter Methods

Warning: This section describes the slaughter process in detail that may be upsetting for some readers.

Once they are full size, the chickens are sent back to the integrator for slaughter and packing.

The first step at the processing plant is to keep the chickens in "lairage," an area for the chickens to "rest" in preparation for slaughter, typically for up to three hours (Saraiva et al., 2020; Johnson, 2018). This is intended to improve the quality of meat by lowering chickens' stress levels (Rodrigues et al., 2017; Puolanne and Immonen, 2014).

When it's time for slaughter, the chickens are removed from the crates they were transported in, shackled by their legs, and hung upside down. According to industry, in order to "minimize

stress," they must be transported to the next stage in under a minute. The chickens are then stunned to make them unconscious before slaughter, though as many advocates know, these processes are not foolproof and errors further increase the suffering associated with slaughter. This stunning is typically done either by wetting the bird's head and electrocuting them or suffocating them with carbon dioxide until they lose consciousness (National Chicken Council, 2013; Johnson, 2018).

After stunning, the chicken's neck is cut to begin the bleeding process, which lasts between 90 seconds and two and a half minutes as gravity drains the blood from the chicken's body. In the "scalding" phase, the chicken's body is plunged into hot water for up to three and a half minutes in order to make feathers easier to remove. This process involves specially designed scalder tanks that pump in clean water to prevent contamination from dirt and excrement. The chicken's body then passes through a series of defeathering machines (Johnson, 2018).

At this stage, the chicken's head and feet are removed and the body is washed. The body is then attached to another set of shackles and the evisceration process begins. At this stage, a worker or machine cuts the body open and begins removing the internal organs. Edible organs, collectively called giblets, are removed and stored separately so they can be packaged and sold. In the U.S., sometimes workers or machines will spray the inside of chickens' bodies with chlorine to prevent contamination from blood or fecal matter (USDA FSIS, 2021; Micciche et al., 2018; Johnson, 2018).

The body is then chilled to slow decomposition. Water chilling is common in the U.S. and is carried out using a freshwater system, often with additives like chlorine and citric acid to minimize contamination. This process can take 90 minutes to three hours. In Brazil, Europe, and Canada, air chilling is more common, and involves placing the bodies in temperature-controlled chilling rooms, which can take much longer to chill the bodies to the desired temperature of 40°F (4°C; Johnson, 2018; Belk et al., 2021).

The Tournament System For Growers

One of the hallmarks of the broiler industry is the tournament system. In the previous section, we noted that DOCs are sent to grow-out farms until they are large enough for slaughter, which are responsible for 99.5% of the broiler industry's chickens (USDA ERS, 2022). Although they are raised by these growers, this is on a contract basis—the chickens are still the property of the integrators, who maintain considerable control over key factors that affect the chicks' growth, such as medical care and feed, as well as setting requirements for the specific "housing, equipment, utilities, and labor" the growers use (USDA, 2022). The growers are responsible for meeting all of these standards and requirements.

Once the chickens from several growers are large enough for slaughter, the integrator will compare the weights of the chickens and award a premium to the growers of the largest chickens while also deducting pay from the growers of smaller chickens: this is the tournament

system. In other words, the farmers who work as growers are required to follow integrators' specifications, but even doing so perfectly, they may still get paid less than neighboring farmers.

Integrators like Tyson save money themselves by outsourcing many of their costs to growers. They are also able to treat growers much like employees, but do not have to provide health care or other benefits (Laninga, 2022). Both through this tournament-style competition and by making growers go into hundreds of thousands of dollars in debt for expensive capital like new ventilation systems in order to continue getting contracts, integrators are able to save money. Integrators have historically made these demands in an effort to increase their profits. Growers have little choice but to keep working with the integrator in the hope of earning enough money to pay off their debt.

As noted above, chicken farmers often have little choice over which companies they contract with because there are often only one or two broiler companies in a given area. Moreover, farmers who may wish to switch to another industry that does not rely on animal exploitation are often in too much debt to make such a change, a problem that is only made worse by the highly specialized facilities and equipment required for raising chickens. In other words, integrators also exploit contract farmers under a system that experts have compared to serfdom (Equitable Growth, 2022; Taylor & Domina, 2010). Activists and legislators interested in strengthening enforcement of the Packers and Stockyards Act—which outlines certain competition regulations for the industry—have tried to strengthen antitrust regulations aimed at preventing this exploitative behavior (Kades, 2022). These efforts are still ongoing and animal advocates may be interested in joining this fight to weaken the power of large animal agriculture companies.

How Indirect Subsidies Benefit The Industry

One of the best ways to understand the effects of government policy on the animal agriculture industry, and the broiler industry more specifically, is to look at the impact of a specific piece of legislation. For our example, we'll look at the 1996 Farm Bill: one of the most impactful pieces of U.S. agricultural legislation in decades. Its repercussions can still be felt in the animal agriculture industry today.

One of the key features of the 1996 Farm Bill was the removal of supply control mechanisms (e.g., saving a portion of grains rather than selling everything) that had been in place since the Great Depression. The policymakers who spearheaded this effort were motivated by beliefs in market deregulation and supported by groups like the National Grain and Feed Association, a trade association with members that include giant grain companies and large broiler integrators (National Grain and Feed Association, 2022; Starmer et al., 2006). Excess crops were now sold rather than stored, so more of these products were available and their prices fell. Between 1997 and 2005 this policy led to corn being sold at 23% less than what it cost to produce and soybeans sold at 15% of what they cost to produce (Starmer et al., 2006). These policy changes did not benefit the farmers who grow these crops: over roughly the same period of time, farm incomes fell by more than 15% (Starmer et al., 2006).

Who, then, does this change benefit? The broiler chicken industry, for one. Corn and soybeans are the main feed ingredients for the broiler industry and feed is the industry's biggest cost. So when the 1996 Farm Bill led to price drops in these crops, it meant that the broiler industry saved an average of $1.25 billion *each year* between 1997 and 2005, or more than $750,000 per broiler factory farm (Gurian-Sherman, 2008). In other words, by removing supply control mechanisms in the grain industry, the 1996 Farm Bill saved the broiler industry billions of dollars (13% of their total operating costs) without accounting for a single direct subsidy for that industry (Starmer et al., 2006). Because these aren't direct government payments to the broiler industry, but the industry benefits greatly nonetheless, these savings are said to be a form of *indirect* subsidy. The USDA has noted that "U.S. poultry production has a competitive advantage in the world due to abundant domestic feed resources, primarily soybean meal and corn" (USDA ERS, 2023). This advantage could evaporate without these subsidies.

Multiple Farm Bills, which determine much of U.S. agricultural policy, have been passed since the 1996 law, but the majority of farm subsidies continue to go to crops like corn and soybeans (Schechinger et al., 2023). In order to keep the farms that grow these crops running after the 1996 Farm Bill, the federal government increased subsidies through subsequent policies like temporary direct payments to farmers or partial payments of crop insurance premiums (Starmer et al., 2006; Howard, 2021). The animal agriculture industry now benefits largely from subsidies to crop farmers rather than the smaller amount of subsidies it receives directly (Tong et al., 2019).

The Costs That Broiler Producers Don't Pay

Just as government action can benefit the animal agriculture industry, so too can government *inaction*. The industry can often benefit from *negative externalities*: costs of the production process that companies and consumers don't pay for themselves. Some are economic, some are physical or social harm done. For animal advocates, the most obvious of these costs of animal agriculture is the suffering animals experience, which is "paid" by the animals, not the company or consumer. Thinking of this as a cost of animal agriculture is more than a thought exercise. Animal suffering increases in farm systems that are "optimized" to lower costs to the industry: Battery cages, for example, lower the cost to farmers at the expense of cost to animals.

Another example of a negative externality of the animal agriculture industry is the Gulf of Mexico Dead Zone. This is an area of the Gulf of Mexico that lacks oxygen, making it nearly impossible for marine life to exist there. This region exists due to farm runoff, such as nitrogen from fertilizers, which reach the gulf via the Mississippi River. More than 40% of the nitrogen involved in this pollution comes from crop fertilizers, primarily those used for corn (Union Of Concerned Scientists, 2020; Howard, 2019). Generally, the impact of this pollution is not reflected in the cost of the crop, which means that these crops are cheaper than they would be if the farmers had to pay to prevent or repair the environmental damage they cause. Those cheaper crops benefit animal agriculture, which buys more than 40% of the corn grown in the U.S. for feed

(USDA ERS, 2023). If grain farmers had to pay for the effects of fertilizer runoff, the cost of most animal products would increase.

In addition to indirect effects like the example above, animal agriculture also directly pollutes the soil, water, and even air. For example, 10% of the pollution contributing to the Gulf of Mexico Dead Zone comes directly from animal agriculture (Union Of Concerned Scientists, 2020). Chicken farmers and integrators are often left with more manure than they can dispose of in an environmentally-friendly way. They spread manure over fields, often in amounts exceeding the legal limit, which then seeps into rivers and groundwater (Foodprint, 2023; Eichmann, 2017). Nitrogen, phosphorus, and even antibiotics from chicken farming operations can be found in the soil and water nearby. This also creates toxic conditions in bodies of water like the Chesapeake Bay and contaminates drinking water (Foodprint, 2023). Poultry processing plants also pollute the air and can lead to the spread of zoonotic diseases like bird flu and *Salmonella* (Baskin-Graves et al., 2019). The effects can be seen on a global scale as well: Almost 15% of greenhouse gases pollution comes from the animal agriculture industry, with about 8% of that coming from the chicken and egg industries (FAO, n.d.).

In other words, the environmental and health costs of the poultry industry are innumerable and are not even fully understood yet (Gržinić et al., 2023). Despite claims to the contrary, integrators consistently pollute the environment as a way of saving money. Increasing the amount of environmental oversight these companies are subject to would ultimately increase the costs of their products while also protecting neighboring communities from environmental damage and health effects.

The Future Of Broiler Production Costs

Industry-Identified Risks

U.S.-based, publicly traded companies file annual reports known as 10-Ks with the Securities and Exchange Commission. One of the requirements of these forms is the “Risk Factors” section, in which companies list the biggest risks they face, typically in order of importance (Investor.gov., 2021). This can allow advocates some insight into what different parts of the animal agriculture industry view as the biggest threats to their continued financial success moving forward.

In Tyson’s 2022 10-K, for example, the company identifies a variety of risks to their business. For instance, Tyson notes that its business would suffer if it were to lose one of its major customers like Walmart, which accounted for 18% of Tyson’s 2022 sales. On the production side, Tyson notes the potential for harm in the case of price fluctuations for raw materials like feed because of how important these inputs are to its business model.

Another of Tyson’s primary concerns relates to labor, which is often one of the highest costs for giant animal agriculture corporations. In particular, Tyson identifies employee turnover, higher wages, and the potential difficulty of attracting new talent as threats to its current costs of

production, noting that "Labor shortages and increased turnover rates within our team members have led to and could in the future lead to increased costs…and could negatively affect our ability to efficiently operate" (Tyson Foods, Inc., 2022, p. 14). Relatedly, the company also notes that about 30% of its workforce is unionized, and that strikes or other union-related activities could raise production costs. In addition to employee labor factors, Tyson also points out the importance of contract farmers to their business model, noting that a loss of these contracts to competitors or as a result of poor relationships would increase costs.

Tyson also identifies legal and regulatory threats to their current cost of production. Among these are limits on greenhouse gas emissions, sustainability initiatives like bans on single use plastics, and actions to increase the cost of power sources like diesel and electricity. Increases in energy and compliance costs could ultimately be very expensive and require significant changes to the company's production facilities and broader business strategies. Put more directly, "The effects of climate change and legal or regulatory initiatives to address climate change could have a long-term adverse impact on [Tyson's] business and results of operations" (Tyson Foods, Inc., 2022, p. 16). The company also identifies various types of litigation, including environmental lawsuits, as ongoing threats.

Pilgrim's Pride echoes many of these risks in their filings. Additionally, it notes a potential threat if consumer preferences change, specifically noting that demand for plant-based protein could grow as a result of "concerns related to human health, climate change, resource conservation and animal welfare" (Pilgrim's Pride 2022 10-K, p. 13). Pilgrim's Pride also fears that its costs could increase if the public and regulators pay more attention to how their food is produced or the environmental and social harms the meat industry causes (Pilgrim's Pride 2022 10-K).

Both Tyson and Pilgrim's Pride note the risk of future outbreaks of disease, including highly pathogenic avian influenza (HPAI). In 2022, more than 38 million birds on poultry farms were killed due to HPAI—mostly by producers "depopulating" the birds rather than directly from the disease. The outbreak affected 220 producers across 35 states and cost the poultry industry over $800 million. A similar outbreak in 2015 resulted in over 50 million birds killed and $1.3 billion losses for the industry (USDA APHIS, 2023; Animal Welfare Institute, 2022). These outbreaks are made worse by the conditions in factory farms, including poor immune systems as a result of stress and closely-packed, genetically similar flocks, though this was not a point highlighted by Tyson and Pilgrim's Pride in their filings (Moore et al., 2021; Jacobs, 2022).

Industry Efforts To Further Reduce Production Costs

The broiler industry is already highly advanced and concentrated, so most of the opportunities for cost-cutting are more efficient versions of current practices. Industry experts are largely focused on automation and disease management (Doughman, 2023). Apart from creating new products, Tyson's current research and development efforts are focused on automating its processing facilities and grow-out farms and further refining the genetics of its chickens—likely including improvements to feed conversion ratios (Tyson Foods, Inc., 2022).

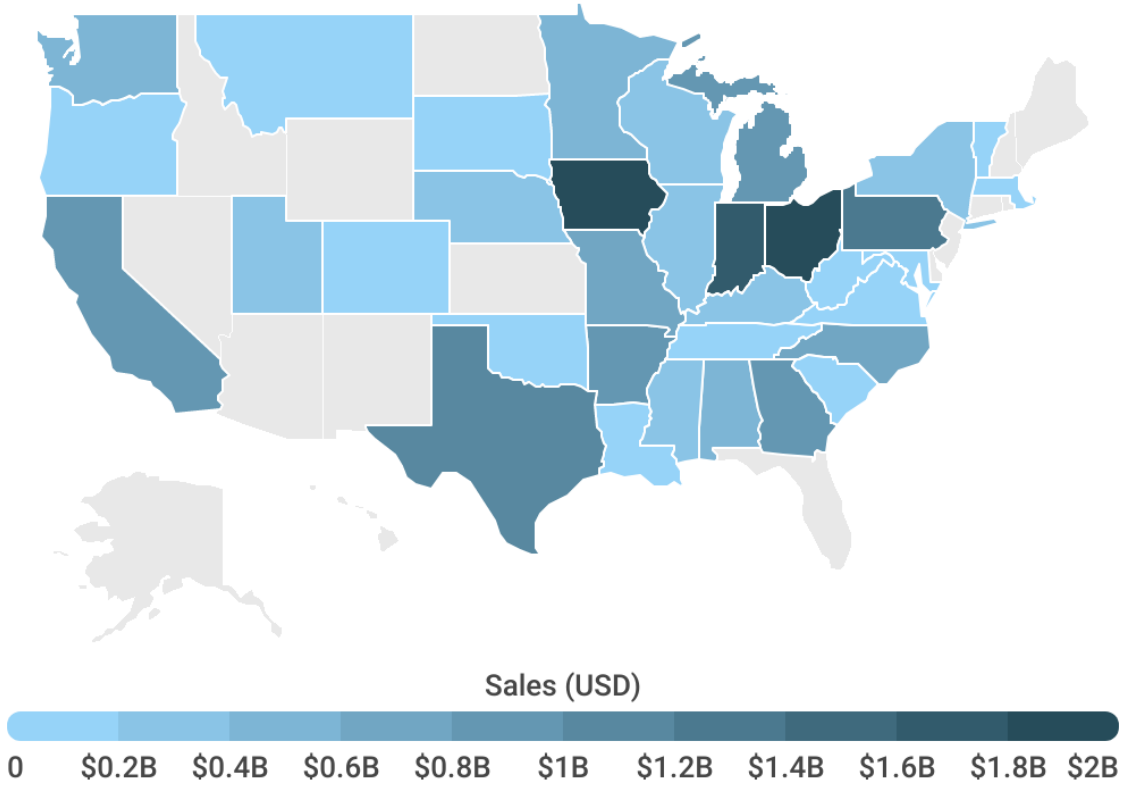
In addition to these efforts, industry publications cite a wide variety of other opportunities to reduce costs, which advocates should be aware of and research further for animal welfare implications: more accurate weighing, improved health tracking, better data on farm conditions like temperature and ammonia levels, and even new approaches to controlling chickens' genetics and microbiomes (Doughman, 2022; Doughman, 2020; Gowda, 2023; Johnson, 2018).

Poultry industries like the broiler and egg industries may also attempt to save money through a process known as *anaerobic digestion,* which captures manure from farms and uses it to create fuel called biogas. Advocates may be familiar with methane digesters, an example of this that has become commonplace on dairy farms (Faunalytics, 2023). This process can provide an additional energy source for farms and may reduce emissions under the proper circumstances, but it can be costly to implement and allows the industry to portray itself as more environmentally friendly than it actually is (EPA, 2023; Rudek & Schwietzke, 2019; Juliani & Pearson, 2020). Anaerobic digestion is also supported by the federal government through tax credits, renewable energy credits, and other benefits (Wisch, 2009).

The U.S. Egg Industry

Measured in layer hens, the largest egg producer in the U.S. and the world is Cal-Maine Foods, which owns 44 million layer hens. With 25 million hens, Rose Acre Farms is the second largest U.S. egg producer and the third largest in the world after Mexico's Proteina Animal (Johnson, 2023, Clements, 2022). In total, the U.S. egg industry owns well over 300 million layer hens, and more than half of those hens are owned by the ten largest egg companies (United Egg Producers, 2023; Johnson, 2023). The U.S. egg industry is not as geographically confined to a single production region as the broiler industry. For example, Texas, California, and Michigan are all top 10 egg producers. Even so, the three biggest producers are Midwestern states. Iowa, Ohio, and Indiana collectively account for nearly 30% of U.S. egg sales (USDA ERS, 2023).

Figure 3. Sales of Eggs Among Top Producing States, 2022



Source: USDA ERS, 2023
Notes: States colored gray combined for $1.3 billion in egg sales in 2022.
7% of national sales are not disclosed to avoid disclosing individual operations.

Over 90% of the eggs produced in the U.S. come from companies that are members of the United Egg Producers (UEP) cooperative (United Egg Producers, 2023). The UEP is a powerful lobbying group for the egg industry and its influence comes in part from the size of its membership. Through a regulatory exemption discussed below, the UEP and similar groups are also essentially able to raise prices legally without being charged with price fixing (Miltner, 2016; USDA, 1985; Pratter, 2016). Although limitations to this power exist, the UEP exerts unrivaled influence over both the market for eggs and industry-friendly legislation and regulations.

In 1922, an exception was made to U.S. antitrust laws in order to allow farmers to form cooperatives that could legally fix prices (Volkin, 1985). This exception came with the passage of the Capper-Volstead Act, which gave small farmers the ability to join together to demand fair prices for their products (Hanley, 2022).

Nowadays there are several agriculture cooperatives that exert large amounts of power over the market for their products, like the UEP. The egg producer cooperative has been accused of anti-competitive practices such as promoting flock culling in order to increase prices for their

products (Pacelle, 2010; Peck, 2015) Because of the size of the cooperative, it essentially sets the standards for the U.S. egg industry through their certification programs (United Egg Producers, 2023; Bolotova, 2019).

The Basics Of Egg Production

While the chicken meat industry focuses on broilers, the egg industry revolves around layer hens. Although the varieties of chickens used in each industry are different, there are many similarities between the two. For instance, both industries use varieties of chickens that technologically advanced genetics companies have specially bred to be as profitable as possible.

Layer hens are carefully bred and their genetics ultimately come from one of a small number of companies who control the market on egg genetics (Fernyhough et al., 2019). White eggs typically come from White Leghorns and brown eggs come from chickens similar to New Hampshire Reds and Barred Plymouth Rocks (Penn State Extension, n.d.). Flocks of breeder chickens, which do not lay eggs for human consumption, are often owned by integrators (Cunningham, 2012). Breeder hens begin laying eggs at about five months old and continue to lay eggs "efficiently" until they are about 21 months old (Penn State Extension, n.d.). The eggs of these chickens, each of which take 23 to 32 hours to produce, are collected daily and sent to a hatchery. Here they are stored in a temperature- and humidity-controlled storage facility for up to a week until they are ready to be incubated. Each of these eggs—intended to hatch a layer hen—is worth about $0.28 to the company (Penn State Extension, n.d.).

The eggs are incubated for 18 days before they are transferred to hatching baskets, where chicks hatch three days later (Penn State Extension, n.d.). The newly-hatched chicks are then processed, which involves vaccinating them and "sexing" them by separating males and females, before they are sent to a grow-out farm within 48 hours of hatching (Penn State Extension, n.d.). These chicks are known as "pullets" and are worth between $0.65 and $0.75. Pullets are kept in a carefully controlled environment where they receive less than 10 hours of artificial light per day in order to prevent premature egg laying (Ockert, 2019). After 18 weeks, pullets are transferred to a layer farm or facility. At this stage, white egg pullets are worth $3.50 and brown egg pullets are worth $3.75 (Penn State Extension, n.d.).

At the layer facility, hens receive 14-16 hours of artificial light each day (Ockert, 2019; Penn State Extension, n.d.). They also receive feed that is similar in composition to the feed or broiler chickens, but with additional vitamins and minerals such as calcium and phosphorus to ensure that their eggs have strong shells (Penn State Extension, n.d.; Bar et al., 2002; Decision Innovation Solutions, 2016). Hens generally lay eggs between the ages of 20 and 60-65 weeks, producing an average of about 275 eggs per year for white egg hens and 260 eggs per year for brown egg hens.

Machines automatically collect, wash, grade, and sort eggs for packaging (Penn State Extension, n.d.). Most eggs in the U.S. are washed according to specific USDA regulations and

are stored in temperature-controlled containers (USDA, n.d.; Grigsby et al., 2018). In addition to conventional shell or table eggs, about 30% of eggs are cracked by processors and sold in liquid, frozen, or other forms (Cal-Maine Foods, Inc., 2023). These eggs are commonly referred to as "breakers" and are frequently sold to the food service industry.

Cost-Cutting At The Expense Of Animal Welfare: Forced Molting, Culling of Male Chickens, and Beak Trimming

The egg companies save as much money as possible by trying to keep layer hens producing at a profitable rate for as long as possible. Farmers may force flocks of layer hens to molt after their initial laying cycle. Historically, this involves providing limited to no feed for up to two weeks, which causes the hens to lose and regrow their feathers and to stop producing eggs. When proper feeding resumes, the hens begin laying eggs again, often at higher rates and of larger sizes (Wigley, 2013).

Forced molting via feed withdrawal is banned in places like the European Union and India, but remains legal in the U.S. In 2006, the United Egg Producers, whose affiliates produce the vast majority of the eggs in the U.S., banned feed withdrawal molt methods, but alternative methods of inducing molting are still employed (Wigley, 2013; United Egg Producers, 2017). These methods include the use of low-nutrient feed and high-zinc diets, both of which disrupt the physiology of hens in order to induce molting (Gha et al., 2022; Khan et al., 2011). The industry generally aims to reduce hens' weights by up to 30% during the molting process in order to obtain the best post-molt performance (Butcher & Miles, 2019; Golden, et al., 2008). Although molting is a natural occurrence for hens in autumn, forced molting can cause significant stress to hens and weaken their immune systems, making them more susceptible to infections (Wigley, 2013).

Hens that no longer lay eggs at a profitable rate are considered "spent" and are sent to slaughter (Fan & Wu, 2022; Pacelle, 2015). The "spent" hens are then replaced with younger hens. Inducing molting cycles stimulates egg production at a profitable rate for egg companies by forcing hens to produce eggs at a higher rate than they naturally would, and also limits the cost of replacing the "spent" hens with pullets.

Animal advocates often discuss the fate of male chicks in the egg industry. Layer chickens are specially bred to produce eggs, while broiler chickens are specially bred for their meat. Because male layer chicks do not produce meat as efficiently as companies want, and because having separate broiler facilities requires significant capital investments, it is cheaper for egg companies to kill male chicks shortly after they hatch than to raise them for meat. In the U.S., the most common way of culling male chicks is maceration, which uses a machine to grind up fully-conscious chicks, but alternative methods such as gassing are also used (Wang et al., 2021; Animal Equality, 2021).

Because the U.S. egg industry currently has more than 300 million layer hens, a similar number of male chicks must have been culled in recent years (Wang et al., 2021). Each year, egg

companies spend over $70 million to incubate and separate eggs containing male chicks.. If these chicks were female, the egg industry would ultimately earn an additional $440 million annually (Doughman, 2020; Business Wire, 2016).

When chickens are kept in conditions like those of factory farms and are prevented from engaging in natural behaviors, they become stressed, which can lead to behaviors like violent pecking and even cannibalism. One way companies in the broiler chicken, breeder, and especially layer hen industries attempt to manage this response is through beak trimming, the removal of up to half of a chicken's beak (Cheng, 2010). Even though it costs companies money to debeak chickens, the practice ultimately saves them money by limiting stress-induced violence and death among flocks.

If they were not born into industrial operations, chicks would spend large amounts of time resting in darkness under their mothers. Natural practices such as this can lead to reduced pecking. Because it is unlikely that major chicken, egg, and breeder companies would allow for natural behaviors, some are experimenting with techniques that seek to replicate these conditions, like brooding simulators and playing recordings of mother hens' calls. Improving housing conditions by ensuring there is plenty of suitable foraging material for chickens, reducing the number of chickens housed together, ensuring proper nutrition, and otherwise enriching the environment can all also reduce the amount of pecking (Nicol, 2018; Epp, 2019).

There is already some precedent for ending the practice of beak trimming. Several European countries have banned the practice—including Norway, which banned it decades ago in 1974, Sweden, Germany, the Netherlands, and others—but it remains common in the U.S. (Nicol, 2018; Epp, 2019). Beak trimming bans should be paired with requirements for improved living conditions in order to minimize pecking-related welfare concerns.

The Costs of Cage-Free And Free Range

A primary factor influencing the cost of production for eggs is the system in which they are raised. Most eggs in the U.S. come from conventionally-raised layer hens, which typically involves the use of battery cages, but alternatives include eggs from cage-free and so-called “free range” systems. Although these approaches come with some disadvantages inherent to any system that confines animals, they provide chickens with more space than the conventional method and also raise costs for producers.

For decades, feed has accounted for about 60% of egg farmers’ production costs (Sumner et al., 2008; USDA ERS, 2006). In recent years, for example, Cal-Maine has spent between 55% and 63% of its production costs on feed (Cal-Maine Foods, Inc., 2023). Research has also shown that feed costs are considerably higher in cage-free and free range systems because chickens with more space are more active and larger spaces are more difficult to manage with precision (Sumner et al., 2008). Because hens in non-cage systems won’t always lay eggs inside designated nesting boxes, the eggs may be more difficult to collect and can get dirty or contaminated (Sumner et al., 2008; Williams, 2022). As a result, non-cage systems also have

lower egg production rates per chicken and egg quality may be lower. Rates of feather pecking and cannibalism may also be higher in the non-cage systems the industry uses, an issue that is often mitigated with techniques like beak trimming.

Labor costs are higher in non-cage systems as well. Workers on cage farms can oversee three times as many hens. Cage farms also allow for easier automation of egg collection, feeding, and waste management (Sumner et al., 2008).

Another factor contributing to higher costs of production in non-cage systems is housing for the chickens. Cage systems actually cost more money overall to construct than non-cage systems, but they also have much higher densities of chickens. Because non-cage systems hold fewer chickens, farmers must spend more per chicken when constructing housing for cage-free or free range production systems, and also have to spend more to maintain these larger spaces. Because farmers may wish to maintain their laying flocks using pullets who were reared in familiar systems, non-cage farmers may also spend more on pullets than farmers who use cages do (Sumner et al., 2008).

One study of California egg farms found that non-cage farms paid about 55% more for pullets, 17% more for feed, 142% more for housing, and 271% more for labor per dozen eggs compared to cage farms. These contributed to a total cost of about 41% more for non-cage farmers (Sumner et al., 2008). Other studies have suggested costs that are at least 20% higher for non-cage farmers (Sumner et al., 2008; Caputo et al., 2023). These estimates are also for farms that produce non-organic eggs. The production of organic eggs entails an array of additional costs, including expensive organic feed and restrictions on the use of antibiotics and hormones (USDA, n.d.).

Even though chickens have more space to engage in natural behaviors in cage-free and free range systems, there are still important animal welfare issues for advocates to be familiar with. Non-caged layer hens, when confined in large numbers, can have higher mortality rates than caged hens. This is in part the result of attacks within the flock due to stress. Exposure to disease because of difficulties in waste disposal also contributes to higher mortality rates (Sumner et al., 2008). However, stronger regulations around the conditions layer hens are kept in could address these issues. This could further increase the cost of eggs produced using these systems.

Rather than improving the conditions of the chickens on their farms, some processors choose to use labels and advertising tactics that could potentially mislead customers about how the company treats its hens. For example, terms like “natural,” “pasture-raised,” and “humanely raised” have no USDA definitions or associated regulations for hen welfare, which lets egg companies put them on cartons without taking any actions to improve quality of life (Leahy, 2021; Animal Outlook, n.d.; Certified Humane, 2018).

Demand for eggs is inelastic, meaning that the amount of eggs people buy generally doesn’t go down much even as prices rise. In fact, demand for eggs is less sensitive than all other major

types of animal products except milk (Okrent & Alston, 2012). As a result, advocates should not necessarily expect egg consumption to drop significantly even if prices increase. Even still, consumers rate price as the most important factor in their egg purchasing decisions (Caputo et al., 2023). The costs producers must pay for non-cage eggs don't entirely get passed on to consumers in the retail price of eggs, but increased costs of production typically result in some amount of price increase. If major U.S. retailers like Wal-Mart and Kroger follow through on their pledges to go entirely cage-free by 2025, more egg producers will have to meet the comparably expensive requirements of cage-free egg production (Caputo et al., 2023).

Government Support

The U.S. egg industry benefits from many of the same programs that support the U.S. broiler industry because both center around chicken exploitation. This includes indirect subsidies that reduce the cost of feed.

As it has with several other animal agriculture industries, the U.S. federal government created and oversees the egg industry's checkoff program, which charges producers fees that pay for the marketing of egg products both domestically and internationally. This program operates as the American Egg Board and works to promote the interests of the egg industry (USDA AMS, n.d.). The American Egg Board is funded by what is essentially a tax on egg producers. Most producers must pay $0.10 for every case they sell, which typically contains 30 dozen eggs (USDA AMS, n.d.). In recent years, the American Egg Board and its USDA overseers have faced scrutiny as a result of the Board "overstepping its congressional mandate" by attempting to prevent the growth of vegan mayonnaise alternative Just Mayo and for government approval of a $3 million campaign against California's Proposition 2, which aimed to improve layer hen welfare (Mohan, 2016; Wisch, 2009).

The Future Of Egg Production Costs

Industry-Identified Risks

Cal-Maine, the largest U.S. egg producer, has identified several risks to its business that apply to the industry more broadly. Because both rely on the exploitation of chickens, many of these risks parallel those of the broiler industry, such as feed price volatility. Both Cal-Maine and Tyson, the leading U.S. broiler company, point out that their business would suffer dramatically if one of their major customers began purchasing from other companies. In 2023, half of Cal-Maine's sales went to its top three customers, the largest of which was Walmart, which accounted for over a third of the egg giant's sales.

Much like broiler companies, Cal-Maine notes that labor is a major cost and that any increase in employee-related costs, such as wage raises or improved benefits, could cause issues for the company. Other risks include price increases to packaging materials and delivery expenses, which the company notes are already significant costs, litigation, and compliance with any new

government regulations. Like in the broiler industry, the U.S. egg industry is also at risk of outbreaks of diseases such as HPAI.

Finally, Cal-Maine identifies pressure from animal advocates as a key risk that could raise both the company's production costs and its marketing costs. It notes that this risk comes largely in the form of cage-free demands from both advocacy groups and consumers. As more of the company's major customers make cage-free commitments, Cal-Maine must spend more to increase the proportion of its eggs that are cage-free and to maintain more expensive cage-free operations. The market for cage-free eggs remains uncertain due to a combination of higher prices and shifting consumer preferences (Cal-Maine Foods, Inc., 2023).

Industry Attempts To Further Reduce Production Costs

One of the major focuses of the egg industry is the development of cost-effective *in-ovo* sexing. This technique would allow companies to identify which eggs contain male embryos during or before they require incubation (WATT Poultry, 2018; King, 2017; Wang et al., 2021). *In-ovo* sexing would reduce costs and meet the demands of consumers and animal advocacy groups who have called for an end to the maceration of conscious male chicks. Although millions have been invested in this effort to date and the public supports it (Cultivate, 2023), there is currently no scalable solution available (Montgomery, 2023).

The U.S. Aquaculture Industry

The Basics Of Aquaculture

Animal advocates tend to be very familiar with concentrated animal feeding operations (CAFOs), better known as factory farms. When many of us imagine CAFOs, we often picture land animals like chickens, pigs, and cows. Less often do we think of aquatic animals like fishes and shrimp. However, the rise of aquaculture over the past three decades has changed the way "seafood" reaches consumers. Just as land animals used for food were moved into more and more confined settings, so too are aquatic animals being forced into pens, nets, or other systems used for aquaculture. This fish farming approach is beginning to replace traditional capture fishing.

Because of the growing role of aquaculture worldwide, this report will focus on fishes produced using aquaculture methods rather than wild caught fishes. However, capture fishing is still a significant source of seafood worldwide and is responsible for major harms to animals and humans, including many that overlap with the harms of aquaculture and other unique harms such as slavery and ghost fishing (Faunalytics, 2023).

Growth Of The Industry

Through the 1970s, when the broiler chicken industry was completing its shift toward consolidation and integration, aquaculture never accounted for more than 12% of global

seafood production (Bailey and Tran, 2019; National Chicken Council, n.d.). By 2021, however, aquaculture was producing 57% of seafood worldwide, compared to only 43% that came from capture (i.e., wild caught) (Ritchie, 2019; World Bank, 2021a World Bank, 2021b). In fact, aquaculture is the fastest-growing food sector in the world and may be on a trajectory similar to the one that produced the highly-concentrated broiler chicken industry we see today (Bailey and Tran, 2019; Ahmed & Azra, 2022).

In the U.S., the vast majority of domestic fish production still comes from capture, with only 9% of fish coming from aquaculture (World Bank, 2021a; World Bank, 2021b). However, the U.S. is also a leading importer of seafood, with between 70% and 85% of the seafood consumed by the U.S. public coming from imports (NOAA Fisheries, 2022). More than half of this comes from aquaculture (NOAA Fisheries, 2022). In other words, even though the U.S. has not yet made the switch to aquaculture for its domestic production, it still relies on aquaculture for a significant proportion of its seafood. With advocate intervention, it may be possible to prevent the U.S. aquaculture industry from growing.

Worldwide, just as much seafood is exported as all other types of meat combined (FAO, 2022). As global fish consumption becomes increasingly reliant on aquaculture, it is critical for animal advocates to include fishes in their definition of farmed animals.

The recent explosion of aquaculture began in the 1990s, which makes the industry considerably younger than other animal agriculture industries. While advocates working on chicken-related issues must fight against a deeply entrenched, highly industrialized industry, advocates focusing on aquatic life have an opportunity to intervene before the aquaculture industry becomes dominated by a small number of massive, vertically integrated corporations. Although there are already many large corporations that are highly active in the aquaculture space, the industry has not yet become concentrated to the extent that the broiler industry has.

However, the concentration process has already begun. For example, Cargill, which is simultaneously one of the world's largest meat processors and largest grain traders, spent over a billion dollars to acquire a Norwegian fish feed manufacturer in 2015 (Bunge & Chopping, 2015). This could enable Cargill to get a foothold in the aquaculture industry in its quest to dominate yet another industry. One of the largest seafood companies in the world is Japan's Maruha Nichiro, which had sales of about $7 billion in 2022 (IntraFish, 2023, Maruha Nichiro Corporation, 2023). By contrast, JBS, the world's largest meat processor, had revenues of nearly $70 billion in 2022 (JBS, 2023). In other words, companies in the seafood industry, while large, are still many times smaller than the largest meat companies (Bailey and Tran, 2019). However, if the aquaculture industry is able to get more concentrated in the future, these companies would be on a similar path to growth as broiler chicken companies.

While some sectors of animal agriculture, like the broiler chicken industry, are working with very similar animals regardless of company and location, the aquaculture industry is much more diverse and therefore has many more factors to consider. This is one of the primary reasons why the industry has not yet become highly concentrated. There are only a small number of

chicken varieties used in the broiler industry (Elfick, 2011). This allows integrators to be as efficient as possible in their production, selecting the feed and conditions based on the specific variety of chicken. By contrast, the aquaculture industry raised at least 334 species of finfishes, 77 species of mollusks, and 48 species of crustaceans in 2021 alone (FAO, 2023b; Bailey and Tran, 2019). Most broiler chickens are raised in standardized settings determined by the integrators. By contrast, aquaculture takes place in "earthen ponds, above-ground tanks, raceways, cages, and pens in freshwater, brackish water, and seawater" (Bailey and Tran, 2019). In addition, some types of fishes, like salmon and tuna, are carnivorous, while others, like tilapia and carp, are not.

Species Diversity

Aquaculture includes diverse animals ranging from salmon to octopuses to clams to shrimp, this report will focus on the most common variety of animals used for aquaculture: finfish. In 2021, carps, barbels, and other cyprinids were the finfish harvested in the largest numbers, followed by miscellaneous freshwater fishes (FAO, 2023b). However, other commonly harvested animals like shrimps and prawns warrant additional research in the future.

Aquaculture can be fairly hands-off (i.e., extensive), very carefully controlled (i.e., intensive), or anywhere in between. Extensive aquaculture typically involves little to no management of conditions, like in the case of oysters which are spread along shorelines before being "harvested" (Allen & Steeby, 2011).

By contrast, intensive aquaculture is much more controlled, with features like temperature management and water aeration. Intensive approaches also require farmers to provide feed to supplement naturally occurring sources, often in substantial amounts to nourish systems with high stocking densities (i.e., huge numbers of fishes kept in a small space). Although extensive systems still play a role in the farmed fish industry, global aquaculture production is projected to grow more intensive over the next 25 years (FAO, 2022). Groups like the Food and Agriculture Organization (FAO) of the United Nations are also calling for increases in aquaculture intensification, particularly among small and medium producers, because they believe it will lead to sustainable development (FAO, 2022). As a result, we have chosen to focus largely on the applications to intensive aquaculture rather than extensive aquaculture.

Costs

Aquaculture often requires significant capital investments. In order to even begin an aquaculture operation, a company must obtain land and/or water to use, enclosures for fishes, and specialized equipment. Increases in the prices of any of these factors, or in the ease of obtaining land and water rights or permits, can make it difficult for new aquaculture companies to get started. Operating costs can also be quite high due to the significant amounts of feed,

water, electricity, and other inputs, as well as labor and maintenance. Across aquaculture, all types of costs can vary considerably depending on the species and approach producers use.

One of the primary inputs in the aquaculture production process are young fishes, sometimes called "fish seed." Fish seed is composed of hatchlings, spawns, fry, and fingerlings, which are all fishes in early stages of their lives, before they have reached full maturity (FAO, n.d.).

Another set of inputs for aquaculture is a collection of chemical materials such as fertilizers, antibiotics and medicines, pesticides, and water treatment materials (Ahmed & Azra, 2022). The most important water treatment material is lime, a mineral used to make water less acidic and more tolerable for fishes. Other inputs often include fertilizers, hormones, probiotics, and even antibiotics (Ahmed & Azra, 2022).

The most common approaches to finfish aquaculture in the U.S. are ponds, concrete channels that water continuously flows through (i.e., raceways), and tanks that clean and recirculate water (i.e., recirculating aquaculture systems; USDA NASS, 2018). These approaches all have important differences and are used to raise different types of aquatic life so the financial costs involved in them can vary, but the most costly aspects consistently include capital, feed, labor, fingerlings, energy like electricity, and farm management.

Let's consider a few examples of how costs differ by species and aquaculture approach. Capital, which includes things like real estate and equipment, is the biggest cost associated with farming trout, pond-raised baitfishes like goldfish and minnows, and pond-raised sportfishes like bream and crappies. Capital is also the biggest cost for increasingly popular recirculating aquaculture systems (RAS), which are tanks that resemble above-ground swimming pools and reuse water that has been cleaned. The biggest cost on catfish farms and most largemouth bass farms is feed, discussed in more detail in the following section. For small farms without hatcheries, fingerlings can sometimes be one of the main costs, while labor is a bigger cost for farms raising golden shiners and fathead minnows. Even among farms that raise the same types of fishes, costs can vary quite a bit. For example, on some largemouth bass farms, feed accounted for less than 40% of their costs, but on others, it comprised more than 70% of their costs (Engle et al., 2019).

In other words, aquaculture is a diverse industry and farms can take a variety of approaches, each with different factors affecting their costs of production. Despite this, the same few costs affect aquaculture farms' ability to produce fishes.

Feed

As is the case with chickens and other land animals used for food, feed is the single costliest input in the production of fishes (Bailey and Tran, 2019). Soybean meal accounts for an estimated 27% of feed, followed by corn and wheat middlings—a flour-making byproduct—at 13%, and dried distillers grains—a byproduct from ethanol production—at 12% (Decision Innovation Solutions, 2017). Even though several of the fishes included in this analysis are

carnivorous, such as trout, less than 6% of feed overall was fish meal. Soybeans, corn, and grains are key ingredients in feed for animals raised via aquaculture, just as with chickens.

The aquaculture industry is consistently working toward reducing the amount of fish meal and fish oil in feed to reduce costs. For example, the National Oceanic and Atmospheric Association's Fisheries department has touted the ability of salmon farmers to decrease the amount of fishmeal they use from 70% in the 1980s to only 25% in 2017 (NOAA, n.d.). This switch keeps costs lower for aquaculturists and is being made even though many fish, like salmon, are naturally carnivorous or omnivorous. Instead, supplemental feed rich in amino acids is used to meet the dietary needs of fishes.

The FAO predicts that aquaculture production will nearly double by 2050 (FAO, 2022). However, the fish feed industry must expand production in order to meet the aquaculture industry's growing need for feed (FAO, 2022, p. 120).

The FAO has identified several key characteristics of economically viable alternative proteins for use in fish feed. Without them, it would be difficult for aquaculture to grow. These characteristics include "sustainable production scalable to commercial levels" and "nutritious [feed] with lower environmental and life cycle impacts" (FAO, 2022, pp. 121-122). Demonstrating the unsustainable and environmentally destructive nature of the feed industry could hinder the aquaculture industry's growth.

The Production Process

Advocates can be more effective if they are familiar with the basics of how fishes are farmed. Because the aquaculture industry uses such a wide variety of fishes, its production processes are quite diverse. In addition to the types of water and structures in which they are raised, there are many other factors that differ across and sometimes even within fish species. A brief overview below outlines the production of Atlantic salmon. Salmon produce the most value for aquaculture companies, so their production is one of the most important for advocates to be familiar with (Ahmed & Azra, 2022). With this exception, detailing the numerous aquaculture production methods is outside the scope of this report. However, advocates would benefit from future research that dives deeper into the production processes for various types of aquaculture-raised fishes.

The industrial farming of salmon begins by selecting broodstock (i.e., breeder fish). Salmon begin their lives in freshwater, grow to maturity in saltwater, and return to freshwater to spawn. As a result, broodstock are taken from their saltwater enclosures and moved to freshwater tanks at a hatchery, where eggs are "stripped" (i.e., removed) from females and mixed with milt (i.e., semen) from males to create fertilized eggs.

Once the eggs harden, they are disinfected and kept in temperature-controlled freshwater until they are ready to hatch. After hatching, the fishes pass through a number of life stages with different requirements. For example, newly-hatched fishes called alevins live in dark conditions on artificial spawning beds and subsist off of the yolk sac from their egg, while fry fishes live in

tanks and are able to feed on their own. Variables like light and temperature are manipulated to bring about smoltification, a series of physiological changes that make salmon able to tolerate saltwater. These fish are then transferred to sea cages where they are raised until they are old enough for slaughter (The Fish Site, 2009).

Raising salmon is a complex process, and even the above process is simplified. Salmon production systems are much more standardized and advanced than those for other species. Freshwater fishes, for example, can spend most of their lives in earthen ponds, concrete raceways, naturally occurring lakes, recirculating tanks, or other production systems. Further differences come from the fish variety and capabilities of the production company. In other words, the production process across aquaculture is quite varied and it may be advantageous for advocates to focus on the aquaculture process for specific fishes.

Slaughter Methods

Warning: This section describes the slaughter process in detail that may be upsetting for some readers.

In addition to the vast array of production methods for farmed fishes, slaughter methods can also differ greatly. Because fish welfare is not a top concern for many, the industry is able to avoid spending more money in order to lessen the suffering of fishes immediately before and during slaughter. Instead, aquaculturists primarily use "methods that were developed to optimise convenience, cost and product quality but with little or no consideration for the welfare of the fish" (Lines & Spence, 2014, p. 255). Because the intensive aquaculture industry is still fairly new, so too are its slaughter methods. Combined with the diversity of fishes, this results in a number of approaches to killing fishes with varying degrees of concern or knowledge about welfare.

Fasting of fishes is typically the first step in the slaughter process. This is done to minimize both waste in the water and inside the fishes. Like most factors affecting fish production and slaughter, fasting methods vary based on the type of fishes and temperature of the water. Fasting periods can often be several days long. Fishes are often starved for longer than needed to cut down on waste or because it's more convenient for the farmer to starve a whole group even when some won't be harvested until later (Lines & Spence, 2014).

After the fasting period, fishes are usually "crowded," which involves concentrating them into a small area with either nets or some type of partition. For producers, this makes it easier to collect and transport the fishes. For the fishes, stress and low levels of oxygen resulting from a large number of fishes in a small amount of space can lead to injury and death (Lines & Spence, 2014).

Although some fishes are slaughtered near where they were raised, many farmed fishes, like salmon, tilapia, carp, and catfish, are typically transported to a different location for slaughter to keep operational costs down. It is expensive to incorporate even the smallest amount of welfare concern into the transportation process. One common transport method is "pumping," which is

often used in salmon aquaculture. Pumping uses a large tube to suck the crowded fishes from one location to another, sometimes over a kilometer away. Another frequently-used technique is "brailing," which uses a net to scoop up crowded fishes. Smaller nets are operated by hand, but larger nets, which can hold hundreds of kilograms of fishes, are operated using cranes. Brails are operated with and without water, with dry brails causing even more stress and injury to fishes. Because it does not require large vacuum tubes, brailing is the cheaper option for most producers. Both of these transportation methods can cause significant amounts of stress and injury from netting, machinery, and contact with other fishes. After transport, fishes are often too exhausted to swim properly or stay upright (Lines & Spence, 2014).

After the fishes are at the slaughter site, producers employ a number of different methods for slaughtering and killing the fishes, "few of which would be considered acceptable for other vertebrates" (Lines & Spence, 2014). The most common of these techniques is asphyxiation without stunning. This can be done by leaving fishes in air until they die, which may take several minutes, but it is often more cost effective for producers to keep the fishes in ice slurry so that they cool while asphyxiating (Algers et al., 2009; Lines & Spence, 2014). During this process, the cold and lack of oxygen may paralyze a fish rather than killing it, making it look as though the fish is not suffering when it actually is. In other species, like tilapia, bleeding without stunning is common. Gutting and beheading while a fish is still conscious is also common and fishes may continue to suffer even after these steps. Salmon are sometimes killed using carbon dioxide, but this method causes visible distress and has been banned in Norway (Lines & Spence, 2014).

There are two main types of stunning used in aquaculture, though most fishes are killed without any form of stunning (Lines & Spence, 2014). Electrical stunning, which resembles a method used in the broiler industry, causes fishes to lose consciousness temporarily either by applying electricity to the water or placing them on a conveyor belt equipped with stunning paddles. Percussive stunning strikes a blow to a specific place on the head of fishes and renders them unconscious permanently or for extended periods when done as intended. However, this can be difficult to do properly due to inconsistencies in fish size, as well as the ability of the worker carrying out the stunning. Producers can also purchase automated percussive stunning machines, though this typically requires a consistent size of fish. Percussive stunning is common for salmon, but many other types of farmed fishes have skulls that make this technique less effective. As a result, wet electrical stunning may be a cheaper option for most aquaculturists (Lines & Spence, 2014).

How Aquaculture Could Grow to Resemble "Big Chicken"

Broiler chickens in the U.S. and throughout much of the world are carefully bred to have genetics that result in the best possible product for the industry. In other words, even across the different varieties used in the broiler industry, the chickens are quite similar.

This is not the case in aquaculture. Although breeder companies play a role in the industry, the standardization of genetics has not reached the same scale as in the broiler industry. Some farmed fishes only live in freshwater, others only in saltwater, and some species thrive in

brackish water or require changes over their lifespan. Some fishes are raised in ponds, others in varying types of inland tanks, others in concrete raceways, and others off the coast or even in the open ocean. When it comes to standardization in the aquaculture industry, the range of species is a limiting factor. However, within types of fishes, such as the Atlantic salmon, many industry norms already exist.

Because aquaculturists are exploiting many species rather than just one, it may ultimately make more sense for advocates to think of aquaculture for different species as separate industries, such as the salmon industry or the catfish industry. In the coming years, standardization is more likely to occur within rather than across these fish-specific industries. The aquaculture industry looks to other animal agriculture systems to serve as roadmaps for growth. Experts expect the industry will look for efficiencies wherever possible since "it is well established that the success of aquaculture in terms of increased production is achieved through innovation, productivity improvements, and cost reduction at all levels in the supply chain, from the production stage to the market" (Asche et al., 2018, p. 2).

Feed As A Pathway To Further Vertical Integration and Concentration

While some vertical integration exists in aquaculture, the industry is not nearly as consolidated as some other animal agriculture industries. However, when it comes to large-scale integration and consolidation efforts by would-be industry titans, it seems more like a *when* than an *if*. One of the main ways this could happen is if seafood processing companies acquire their feed suppliers. Companies like Tyson and Cargill have shown how this can benefit meat processors, and because many of the feed ingredients are the same, it's not difficult to envision seafood processors following a similar route.

Waste from seafood processing is also one of the main sources of fishmeal and fish oil, both of which are commonly fed back to fishes, meaning that seafood processors are already well-positioned to become their own feed suppliers (Bailey and Tran, 2019). Although fully vertically integrated companies have not yet taken over the aquaculture space, some large seafood companies, like South Korea's Dongwon Group and Thailand's Charoen Pokhand, are already involved in both feed production and aquaculture (Bailey and Tran, 2019). It's also possible that megacorporations that already dominate the land animal markets, like JBS, Tyson, and WH Group, will take advantage of their already bountiful resources to expand into yet another industry under the animal agriculture umbrella.

Mirroring land animal agriculture, the aquaculture industry is increasingly using crops like soybeans in fish feed. In 2000, nearly a quarter of fish feed came from fishmeal or fish oil. By 2020, as land crops became a more common ingredient, that number was down to about 7% (Fry et al., 2016). While this is a positive development for the fishes typically used for fish meal, it is also a visible sign of the efforts aquaculturists are undertaking to cut costs in their efforts to industrialize and further grow their industry (Asche et al., 2018).

As aquaculture subindustries have matured, they have also become more efficient. For example, feed conversion ratios for Atlantic salmon decreased by nearly a third between the 1980s and early 2000s (Guttormsen, 2002; Asche et al., 2018). Typically, companies now administer feed in the form of specially-formulated pellets that resemble dry dog food, giving them more control over the fishes' nutrition (NOAA, n.d.).

Certifications

In addition to feed approaches, a handful of other routes can also lead to the aquaculture industry resembling "Big Chicken." One of these routes is through product certifications. These certifications are often created for seemingly good reasons, with objectives such as protecting the environment, avoiding harmful effects to humans, and minimizing harm to animals, among other goals. However, they are often expensive and require specific technical expertise to the extent that smaller producers can't obtain them. As a result, companies that already have abundant resources are given a leg up. As consumers call for stricter regulations around seafood and retail stores respond accordingly, certifications will become necessities, allowing big seafood companies to strengthen their grip on the market and keep others out. Some experts fear that this could ultimately lead to oligopolies in aquaculture with a few large corporations controlling the market for seafood products (Bailey and Tran, 2019). As noted in the broiler chicken section, this can result in harm to both humans and non-human animals, such as poor animal welfare practices or undue political influence.

Selective Breeding and Genetic Engineering

Another way for more advanced enterprises to leverage their already existing resources is through selective breeding and genetic engineering. For example, for some species of fishes, there are considerable differences in size between females and males. Companies can manipulate growth cycles with hormones, temperature, and pressure changes to ensure that they only get the larger fishes. They are also experimenting with creating hybrid fishes that are larger and more resilient. Through manipulations to pressure and temperature or with the use of chemicals, aquaculture producers can create sterile fishes known as triploid fishes. Because their bodies do not divert valuable resources to sex organs, these fishes grow larger and are therefore more valuable (Bailey and Tran, 2019). Perhaps the most extreme form of these techniques is genetic engineering, made easier by fully sequenced genomes for varieties of salmon, catfish, and many other aquatic species. Smaller producers will be unable to compete with the products of these efforts, which are often patentable, further protecting the position of already large corporations (Bailey and Tran, 2019).

This parallels the chicken industry. One of the main companies in salmon and trout genetics is AquaGen AS, which was acquired by EW Group in 2008 (AquaGen, n.d.; IntraFish, 2008). Since then EW Group has purchased two companies that specialize in tilapia genetics (GenoMar, n.d.; Fish Farming Expert, 2020). EW Group also owns Aviagen, which accounts for 44% of all broiler breeder sales around the globe, making it the world's biggest broiler breeding

company (USDA APHIS, 2022). In other words, aquaculture is already seeing corporations attempting to replicate the consolidation tactics used in other animal agriculture industries.

The Costs That Aquaculturists Don't Pay

In addition to the obvious animal welfare issues, there are several other concerns that will only grow more prominent as aquaculture continues to expand. These costs are not fully accounted for in the production process of seafood products. Economists call such costs negative externalities because they are paid for by people and animals other than aquaculturists or seafood consumers, both of whom are more directly involved in the industry. Negative externalities can take many forms, including economic, physical, or social harms. If aquaculture companies were made to address these issues, the costs of their production and therefore the products they sell would ultimately increase.

In the U.S., the most common aquaculture techniques use some form of inland confinement system to raise fishes, most of which require massive amounts of freshwater to maintain water quality (USDA NASS, 2018). These include earthen ponds, raceways, which are narrow channels made of concrete, and tanks, which are typically made of metal, plastic, or fiberglass.

13% of farms use recirculating aquaculture systems (RAS), which reduce the environmental impacts of contaminated wastewater by filtering and reusing water. However, the environmental gains of these systems come with clear animal welfare costs as a result of the cramped, enrichment-free conditions, and are harmful to the environment in other ways. They are energy-intensive and also produce nitrous oxide (N_2O), a greenhouse gas with a warming potential more than 300 times that of CO_2 (Chen & Gao, 2023; Hu et al., 2012; Little et al., 2018).

Other systems are no better, producing a different set of problems. Ponds, raceways, non-recirculating systems, and other inland confinement systems all risk contaminating groundwater and nearby bodies of water. Wastewater may contain fish waste, uneaten feed, antibiotics, high levels of salt water, and more harmful pollutants that can contaminate water and soil, affecting both humans and non-human animals.

In both inland systems and marine systems, such as floating pens, disease and parasites are major concerns. These can result from water quality issues and from the stress placed on crowded fishes. Aquaculturists often deal with these problems through the liberal use of antibiotics, which can easily escape the system and contaminate the environment (Bailey and Tran, 2019). Because many aquaculture systems are located in oceans, rivers, and lakes, parasites like sea lice can quickly spread from enclosures to wild fishes (Krkošek et al., 2005).

Another major issue with aquaculture is the potential for farmed fishes, especially salmon, to escape from their enclosures and enter into larger bodies of water, competing with wild aquatic life. For example, in 2017 more than 250,000 non-native Atlantic salmon escaped from a pen enclosure into Washington's Salish Sea (Schoot et al., 2021). Escaped fishes may outcompete

and displace native species, reduce the adaptations of wild fish, become invasive species, or otherwise negatively impact the ecosystem (SeaChoice, n.d.; Thorstad et al., 2008).

The growth of aquaculture brings substantial risk to marginalized humans as well. Contrary to the overall trend of the meat industry, seafood is more often produced in the Global South and exported to the Global North. As corporate interests intensify the production of the highest-value aquatic creatures, like salmon and shrimp, some experts fear that a new, international system of producer exploitation will emerge (Bailey & Tran, 2019). Similar to the network of small growers that characterizes the broiler industry, powerful aquaculture corporations may contract a system of small-scale farmers in the Global South who become dependent upon the big companies for their livelihoods. As the existence of garment industry sweatshops makes clear, this could result in a considerable amount of human exploitation across borders.

Cutting Costs

Economies of Scale

Like many other animal agriculture producers, aquaculture farms also exhibit *economies of scale*, essentially meaning that the cost per fish goes down as the number of fishes increases (Kenton, 2022). For example, an aquaculture farm may spend $1.00 per fish when it produces 1,000 fishes but only $0.50 per fish when it produces 5,000 fishes. This encourages smaller farmers to grow larger and also means that bigger aquaculture companies are better positioned to dominate the market, driving smaller producers out of the business or absorbing them. It may also enable producers using RAS, for which capital expenditures like tank construction are the greatest cost, to decrease per-fish costs (Engle et al., 2019).

Government Support

The U.S. government is taking steps to support the expansion of the country's aquaculture industry. In fact, the White House's National Science and Technology Council has said that "a dramatic increase in domestic aquaculture is needed to complement well-managed wild-harvest fisheries, drive down the trade deficit in seafood, and meet the growing demand for seafood" (NSTC Subcommittee on Aquaculture, 2022).

Several federal agencies are also collaborating on a number of research and development initiatives and grant programs to expand U.S. aquaculture production, including the Departments of Commerce, the Interior, Agriculture, Education, and Defense, as well as the Environmental Protection Agency. In other words, the U.S. government views aquaculture as a key industry to funnel money into in the coming years. This is partly supported with research led by government agencies or by federally-funded land-grant universities.

Another way the U.S. government may spur growth in its domestic aquaculture industry is by reducing regulations. Some experts have argued that U.S. aquaculture producers are faced with a particularly complex web of federal and state regulations that make doing business more difficult (Engle & Stone, 2013). Laxer regulatory environments in other countries, particularly

those in Asia, around the use of antibiotics and other drugs and chemicals may contribute to their success in the global aquaculture market. In other cases in the U.S., certifications called for by non-governmental organizations (NGOs) have increased costs. Some countries with economies and regulatory environments more similar to the U.S., such as Norway, have much more streamlined and efficient permitting and regulatory processes. These could serve as a model for the U.S. in the future (Engle & Stone, 2013).

Criticisms may be leveled at current waste treatment policies, which some industry proponents view as overly burdensome (Engle & Stone, 2013). While many animal and environmental advocates would undoubtedly wish to increase the regulations on aquaculture, particularly around animal welfare and environmental impacts, it is likely that promoters of aquaculture will seek to dismantle much of the existing regulatory environment surrounding the industry. As the aquaculture industry grows, it may follow other animal agriculture industries and invest more money in lobbying in an effort to influence government policies.

The Future Of Aquaculture Costs

The aquaculture industry in the U.S. is still quite small compared to many other countries around the world, ranked 18th in total production (NOAA, 2022). However, this could change in coming years, especially if the U.S. government continues to express interest in growing the industry. Currently, catfish production makes up the largest segment of U.S. aquaculture, with the industry centered in Mississippi, Arkansas, and Alabama (Fantom, 2023).

Aquaculture in the U.S. and around the world could also benefit from a variety of new technologies and efficiencies. Some of these are related to feed, including nutritional refinement, alternative or cheaper ingredients, and more efficient feed conversion ratios. New products and improved technology for processes like deboning will also lead to industry growth (Asche et al., 2018).

Brazil

In many ways, the Brazilian animal agriculture industry greatly resembles that of the U.S. For example, both countries are top-three global producers of both corn and soybeans, and both countries have world-leading, vertically integrated broiler industries (FAO, 2023a; Valdes et al., 2015). As a result, much of the information discussed in the U.S. sections above applies to Brazil as well. In this section of the report, we outline some additional facets of the Brazilian broiler and egg industries. Brazil does not have a large aquaculture sector, nor does it rank among the world's largest importers of fish products, so we will not discuss its role in the aquaculture industry here.

In order to understand Brazil's current role as a global leader in animal agriculture, we must understand how Brazilian companies grew to their current size. A brief overview of this history provides background for Brazil and shows how other countries might try to replicate Brazil's rise. Although the Brazilian government does not spend a lot of money on agriculture compared to

many other countries, its actions in recent decades helped spark a boom in its animal agriculture industry. For example, neoliberal policy approaches such as privatization and deregulation took hold in the 1990s and led to the absorption of small farms by larger operations (Scott & Vigo, 2021). Also during this time, increasingly urban consumers began to purchase integrator-produced chickens from grocery stores rather than chicken farms (Scott & Vigo, 2021). Crop subsidies and projects like the creation of transportation infrastructure for soybeans indirectly assisted the meat industry by reducing the price of feed (Scott & Vigo, 2021; Fearnside, 2001; Howard, 2019; da Silva, 2021). Brazilian integrators also began to grow more efficient in the 1990s, incorporating new technologies into the production process and finding uses for what had previously been waste products. In the 2000s, an influx of government funds would support the growth of major domestic poultry producers.

JBS And Direct Government Support

Before diving into Brazil's broiler chicken and egg industries, we need to start with a discussion of JBS, the Brazilian megacorporation that controls a huge proportion of the poultry, beef, and pork industries in the U.S. and other countries. With its multi-industry dominance, it is a massive force shaping both Brazilian and global animal agriculture.

JBS, the world's largest meat processor, and other companies have greatly benefited from the direct financial support of the Brazilian government (Howard, 2019). JBS is one of a handful of agricultural companies that have benefited from Brazil's "national champions" development strategy. Beginning around the early 2000s, the Brazilian government began identifying domestic corporations that were or could become global leaders in industries like agriculture, telecommunications, mining, and oil. These "national champions" became privileged companies, receiving various financial benefits from the government.

One of the primary ways that JBS and other "national champion" companies receive financial support is through BNDES, a development bank owned by the government of Brazil, which provides loans and other forms of financing to Brazilian companies. In 2007, BNDES became a minority shareholder in JBS, meaning that the Brazilian government indirectly became a partial owner of the meat processor (Howard, 2019). As of August 2023, BNDES owned 21% of JBS and it has previously owned as much as 31% (JBS, 2023a; Degen, 2012). In addition to the direct financial investment JBS received from BNDES becoming a shareholder, the company also received access to loans with lower interest rates than they would receive from non-BNDES lenders (Degen, 2012). In 2007, JBS acquired Swift, one of the largest meat processors in the U.S. and a company more than five times larger than JBS (Bell & Ross, 2008). JBS would go on to acquire two more large U.S. meat processing operations: Smithfield's beef segment and Pilgrim's Pride, in 2008 and 2009, respectively. Each of these deals was made possible with funding from BNDES (Freitas et al., 2017).

Government investigators have alleged that JBS received approvals for particularly risky loans at especially fast rates (Howard, 2019). In 2017, executives at JBS admitted that the company

had given at least $220 million in bribes to politicians in order to grow as fast as it did (Freitas et al., 2017).

In addition to JBS, BNDES also finances Marfrig and BRF, two other Brazilian meat processing giants (Howard, 2019; Mandl et al., 2019). In 2018, Marfrig purchased the fourth largest beef processor in the U.S., National Beef Packing (Reuters, 2021; Howard, 2019). All of this support from the Brazilian government, both legal and illegal, has allowed large Brazilian meat processors to grow both more cheaply and more quickly than they otherwise would have, taking over a huge proportion of the global industry as a result.

The Brazilian Broiler Chicken Industry

As recently as 2022, Brazil was the third largest broiler industry in the world behind, but has recently surpassed China to become second, now only trailing the U.S. (Aquino, 2022, Aquino, 2023). The country's new second-place position is emblematic of the rapid growth of the domestic broiler industry. In only the last two decades, Brazil has more than doubled the number of chickens it slaughters (Aquino, 2023).

Like its U.S. counterpart, the Brazilian broiler chicken industry has used market concentration and vertical integration to expand its reach around the globe. Brazil is the world's largest exporter of chicken and is responsible for more than a third of the world's chicken exports (Aquino, 2023). One out of every eight chickens Brazil exports goes to China. Other major customers include the United Arab Emirates, Japan, Saudi Arabia, and South Africa (Aquino, 2023). Even other global leaders' production can be tied back to Brazil. As noted above, for instance, Brazil's JBS owns the second largest chicken producer in the U.S., Pilgrim's Pride.

Brazil's broiler industry is heavily concentrated in the South and Southeast regions, particularly in the states of Paraná (30% of production), Rio Grande do Sul (12%), São Paulo (12%), and Santa Catarina (10%; Scott & Vigo, 2021). Compared to the rest of the country, these states have historically had more access to feed crops, urban consumers, and ports for exports. In total, Brazil's Southern region slaughters nearly 60% of the country's broiler chickens, followed by the Southeast, which is responsible for 21% of Brazil's broiler chicken production (Embrapa, 2023). Since the 1990s, the industry has also expanded into the Center-West region, which now slaughters 15% of the country's broiler chickens following increased production of feed crops in this region (Valdes et al., 2015; Embrapa, 2023).

Industry Concentration

The Brazilian poultry industry is extremely concentrated, with only two companies, JBS and BRF (formerly known as Brasil Foods), contributing nearly 50% of the country's overall production and 70% of its exports. This is a massive increase from only a decade ago, when the top *four* producers accounted for 38% of overall production (Scott & Vigo, 2021; Klein & Luna, 2022; Valdes et al., 2015). Both JBS and BRF have received support from BNDES, which significantly contributed to their growth (de Souza & Terra, 2021). However, there are multiple routes to

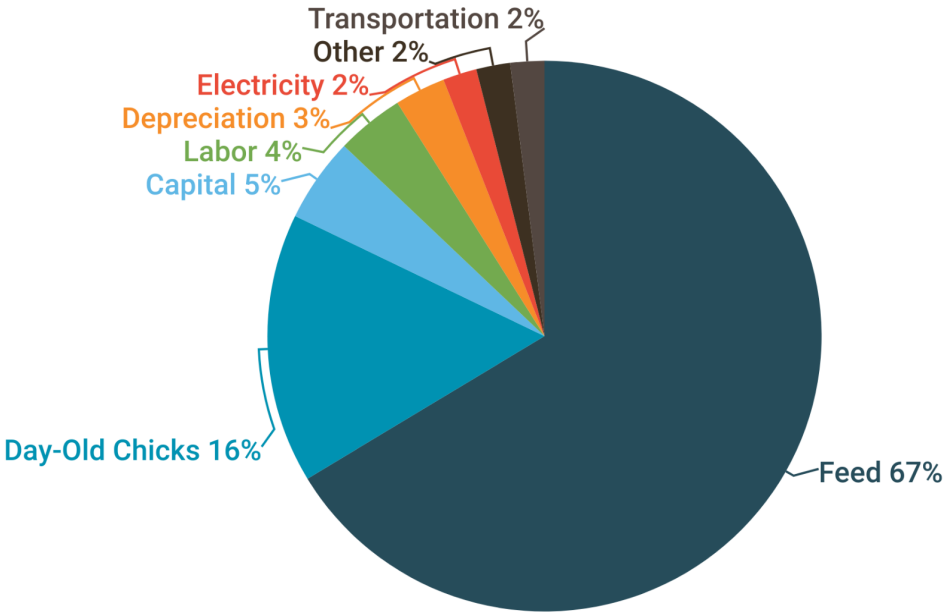
industry dominance. JBS has also grown through the aggressive acquisition of companies around the world, especially in North America. BRF, by contrast, was formed in 2009 through the merger of two already established companies, Pedigão and Sadia (The Poultry Site, 2009; Ruiz, 2016). It has focused on buying small companies around the world and targeting the halal market in the Middle East (Sharma & Schlesinger, 2017; Ruiz, 2016).

Like their industry-dominating counterparts in the U.S., JBS and BRF can exert significant influence over the chicken market in their own country and abroad as a result of industry concentration. This degree of concentration can also enable companies to engage in illegal activities like price fixing. For example, in 2020, the Brazilian government launched a probe into JBS and BRF after their executives said they planned to raise chicken prices (Brito, 2020).

Feed And Other Production Costs

In Brazil, like the U.S. and other countries, feed is the largest expense the broiler industry faces. Two-thirds (67%) of the cost of production comes from feed, with an additional 16% coming from the cost of day-old chickens purchased from hatcheries. The remaining 17% comes from a variety of other costs such as labor, and electricity, as shown in Figure 4 (Embrapa, 2023; Aquino, 2023).

Figure 4. Brazilian Broiler Chicken Production Costs



Source: Embrapa, 2023

Feed costs are often lower for large operations and for those in closer proximity to feed crop production areas (Valdes et al., 2015). These differences can be dramatic when considered on a per-bird basis. Data from 2010 shows that the per-bird costs of farms with 25,000 or more

chickens were less than half of the costs of farms with fewer than 19,000 chickens. Fewer than half of the smaller-sized farms were profitable, compared to over 90% of the larger farms (Valdes et al., 2015).

Feed conversion ratios, which measure the amount of feed needed to produce more flesh for meat on a chicken's body, currently appear to be slightly less efficient in Brazil compared to the U.S. broiler industry (National Chicken Council, 2022; Best, 2011; da Silva Lima et al., 2019).

Contract Farmers

About 85% of the broiler chickens produced in Brazil come from contract growers, many of whom are subject to something resembling the tournament system, paralleling the U.S. industry (Valdes et al., 2015).

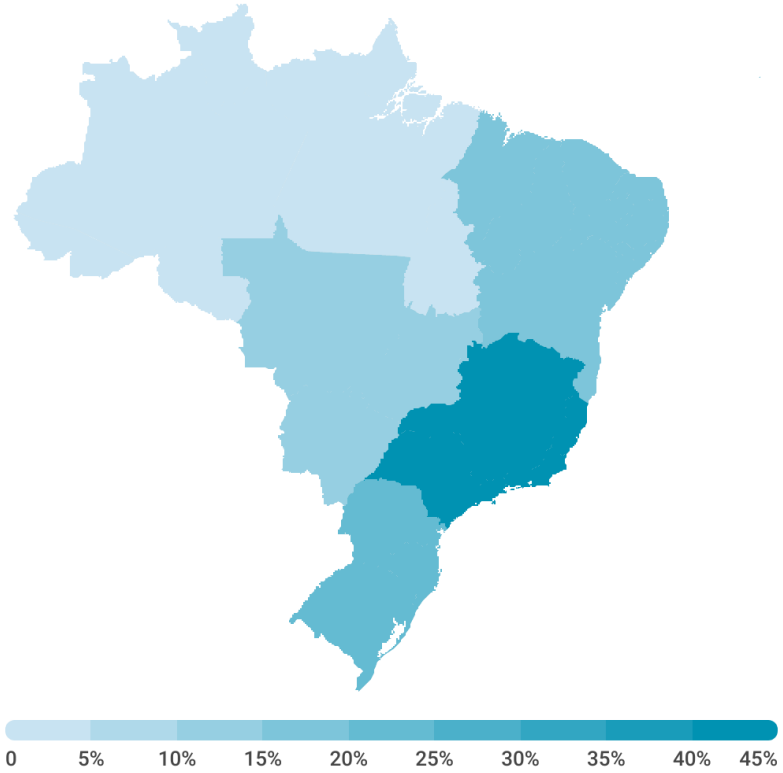
Because the Brazilian poultry industry features a system of contract growers similar to that found in the U.S., integrators are able to externalize much of their costs on to smaller contract farmers. This system can be highly exploitative, with debt-ridden growers sometimes spending more on running their operations than they are paid by integrators (Sharma & Schlesinger, 2017). JBS and BRF have both also contracted with producers who were later added to the government's list of companies who used slave labor (Repórter Brasil, 2016; Human Trafficking Search, 2015).

The Brazilian Egg Industry

In 2022, Brazil was the fifth biggest egg producer worldwide, with more than 99% of those eggs consumed domestically (Maciel et al., 2023; FAO, 2023a). Egg consumption has rapidly increased in recent years even as prices have risen (Lanza & Mendes, 2022; Aquino, 2023).

The geographic distributions of egg production and broiler chicken production are considerably different. The egg industry is concentrated in the country's Southeast region, which produces 45% of Brazil's eggs. By contrast, Southeastern farmers raise fewer than a quarter of the country's broiler chickens. The broiler industry is concentrated in the South, but only 20% of Brazilian eggs are produced in this region. Nearly 20% of Brazil's eggs come from the Northeast, but this region is only a minor contributor to broiler chicken production totals (Embrapa, 2023). Advocates may be more impactful by working with regional governments or labor groups on issues related to the specific types of chickens—broilers or layers—that are more common in their regions.

Figure 5. Proportion of Brazilian Egg Production by Region, 2020



Source: Embrapa, 2023

If the Brazilian egg industry wants to begin exporting larger quantities, it will likely need to improve the welfare of its chickens in order to meet stricter requirements in places like the European Union (Poult, 2012; Busby, 2019; Global Ag Media, 2023). Currently, about 95% of Brazilian eggs come from caged chickens, with about two-thirds raised in a staggered cage arrangement and about a third in a stacked cage arrangement (Maciel et al., 2023).

Despite a massive difference in exports, the Brazilian and U.S. egg industries follow a similar production process, beginning with feed mills and breeders and ending with factory farms and the slaughter of “spent” layer hens. Similar to its broiler companies, Brazil’s egg companies benefit from access to abundant feed crops like corn and soybeans. The largest of these companies is Grupo Mantiqueira, which owns 11.5 million layer hens, followed by Granja Yabuta and Granja Faria, each of which own 10 million layer hens (Ruiz, 2022). Grupo Mantiqueira has committed to not building any additional conventional cage facilities and to having 2.5 million cage-free layer hens by 2025 (Humane Society International, 2022).

As climate change drives up temperatures in Brazil, particularly in the chicken and egg regions of the country, costs will increase for egg producers. For those with climate controls, energy

costs will increase, while producers without climatized systems will need to upgrade in order to prevent deaths from overheating (Lamarca et al., 2018).

Need For Additional Research

Possibly because of its minimal role in the international market, little English-language research has been done on the Brazilian egg industry, particularly when compared to the domestic broiler and U.S. egg industries. The Brazilian Ministry of Agriculture's research wing, Embrapa, provides some English translations of its publications, but information is still limited and does not include advocates' perspectives. Funders should consider opportunities for additional research into this industry, which exploits tens of millions of chickens each year. Moreover, the amount of animal suffering the Brazilian egg industry causes could increase even more in the future if the country begins to look outside its borders for new customers.

Negative Externalities Of Both Industries

Like their counterparts in the U.S., the Brazilian broiler and egg industries are often able to avoid paying for the harms of their production processes. In fact, many of these are the same across both countries, such as environmental degradation and worker exploitation, as well as animal suffering.

Unique to Brazil is the destruction that the broiler and egg industries wreak on the Amazon rainforest in northwestern Brazil. In addition to the well-documented impacts of the beef industry on the Amazon, soybeans for chicken feed have also contributed to the rainforest's deforestation (Sharma & Schlesinger, 2017). Before 2008, 30% of Amazon deforestation was due to the soybean industry (Anand, 2022). A few years after a 2006 soy-growing moratorium was implemented in the region, soybean expansion in the Amazon dropped to the low single digits (Rausch, 2021).

Unfortunately, while the moratorium on soy helped the Amazon, it had negative consequences for other parts of the country. The Cerrado, a large savannah region in the eastern and Central-West regions of Brazil, is now being erased by soybean and corn producers in order to feed farmed animals around the world, including chickens and fishes. Over the past few decades, more than half of the native vegetation in the Cerrado has been deforested in order to make room for soybeans (Gibbs et al., 2015; S&P Global Commodity Insights, 2022). While nearly 80% of the Amazon is protected by conservation laws, the same is only true for 20-35% of the Cerrado (S&P Global Commodity Insights, 2022; Phillips, 2020). Deforestation of the region, which is a massive carbon sink, is one of the main contributors to Brazil's greenhouse gas emissions (Phillips, 2020; Reuters in São Paulo, 2022). The expansion of crops used primarily for animal feed has also threatened the livelihoods of traditional communities, including Indigenous people, in Brazil (Lopes et al., 2021; ActionAid, 2017; Resende et al., 2021).

The Future Of Broiler And Egg Production Costs In Brazil

To date, Brazil's chickens have remained free of avian influenza, which has affected chickens in several other countries. Because of this, Brazil's exports have been more attractive to other countries, a trend which could continue in the future (Clements, 2023). If Brazilian flocks remain free of avian influenza, a growing international customer base would likely enable Brazil's broiler industry to continue expanding and intensifying in order to meet increased demand. Animal welfare may grow more important to Brazilian consumers in the coming years, which could lead to higher costs as facilities convert to cage-free systems or otherwise take steps to improve chickens' quality of life (Bonamigo et al., 2012). Brazilian poultry producers may also look to expand their generation of biogas in the future, which can be expensive in the short term but ultimately save them money on waste disposal and potentially on energy costs (McCarthy, 2023; dos Santos et al., 2023).

The Brazilian government has also identified a number of factors that may affect the cost of production in the future. These include environmental regulations, efforts to improve sustainability, and innovations that will reduce water usage. Other considerations include various improvements to animal welfare, automation, and genetics, as well as the discontinuation of antibiotic use (Bassi and da Silva, n.d.; Embrapa, n.d.).

In order to strengthen their trade relationship, China and its major grain trading company COFCO have made significant investments in Brazil's crop agriculture industry, underscoring both the global nature of the animal agriculture industry and its interconnectedness with the global grain trade (Reuters, 2023; Hiratuka, 2022).

China

China is a dominant force in the animal agriculture industry. Its fish production is more than three times greater than the second biggest fish producer, India, and its egg production is more than five times greater than the second biggest producer, the U.S., while it barely trails the U.S. in chicken production (FAO, 2023a).

China enables animal agriculture outside of its borders as well. For example, over 60% of China's chicken imports come from Brazil, followed by the U.S. at 16% (Baych, 2023). Its massive influence can also be seen in the feed industry: in 2020, the country was the second biggest corn producer and fourth biggest soybean producer, while also one of the top two importers of both crops in order to meet its immense need for feed (FAO, 2023a). An accurate portrait of global animal agriculture and its far-reaching consequences requires a thorough understanding of China's role.

The Chinese Broiler Chicken Industry

China is one of the largest producers and consumers of chicken products in the world. Its broiler industry has grown rapidly in recent decades and is continuing to standardize and intensify as it

grows. Despite production that rivals that of the U.S. or Brazil, the Chinese broiler industry is unique in several ways discussed below, including in the types of chickens it uses and the role of the government.

Researchers at the Institute for Agricultural and Trade Policy have identified four stages of development in the Chinese poultry industry. From 1961 to 1978, broiler production was characterized by a slow rate of growth and was mostly composed of backyard farmers. After the government made economic and political reforms, growth increased rapidly from 1979 to 1996. Beginning during this period and accelerating from 1997 to 2009, foreign investments led to increased intensification in the form of factory farming, as well as greater industry concentration and vertical integration. For example, between 1996 and 2011, the number of broiler farms in China decreased by 75%, suggesting that rapid consolidation was taking place. Since 2010, the industry has been in a "restructuring and upgrading" stage, focused largely on improving food safety and continued growth and standardization (Pi et al., 2014).

Several Types Of Chickens

In the U.S., Brazil, and much of the rest of the world, producers raise chickens with white feathers. These "white broilers" are imported into China, where they compete with China's traditional yellow-feathered chicken market. In 2020, the industry estimated that half of all chicken meat in China came from white broilers and about a third from yellow broilers.

Two thirds of yellow broilers, which do not grow as fast or as large as white broilers and tend to cost more, are typically raised by small and medium producers and sold in wet markets (Baych, 2023; Mason & Mano, 2017; Baych, 2022). Chinese consumers generally use yellow and white broilers for different purposes, so they are not perfect substitutes for one another.

In recent years, as live poultry markets close, feed costs increase, and white broiler producers continue to promote slaughtered and chilled chicken meat, small and medium yellow broiler producers have begun to close their facilities. Although the Chinese broiler industry is still a mix of white and yellow broilers, experts predict that white broiler consumption will continue to displace yellow broiler consumption (Baych, 2023). The white broiler supply will come largely from imports and the domestic industry in Shandong Province and Northern China (Inouye, 2020). The decline in yellow broilers will mostly affect the South, especially Guangdong Province.

The remaining 18% of chicken meat consumed in China comes from hybrid "817" chickens and "spent" hens (Inouye, 2020). 817 chickens are a breed specific to China and are a cross between a commercial layer hen and white feather chicken. They make up about 10% of Chinese chicken meat production and grow faster than white broilers, though not as large. The USDA notes that 817 chickens, which are typically sold pre-cooked and are commonly purchased online, are "the right size for the typical three-person Chinese family" (Inouye, 2020). So-called "spent" layer hens make up the remaining 8% of Chinese chicken consumption.

Broiler Chicken Breeding

Breeders refine chicken genetics over several generations. “Pedigree” chickens are bred to produce great grandparent chickens, who are bred to produce grandparent chickens. The next generation, parent chickens, are bred to produce the chicks that are ultimately killed and eaten (USDA APHIS &| Iowa State University, 2022). Chinese companies typically import white-feathered grandparent chickens (Inouye, 2020; Baych, 2023).

In recent years, due to an outbreak of HPAI in several countries with chicken genetics industries, China has limited its imports of grandparent chickens. In order to maintain broiler production, companies utilized forced molting to make grandparent and parent generations produce more chicks. This practice allows Chinese companies to maintain their flock sizes when fewer breeder chickens are available for import (Baych, 2023).

In 2021, China developed its first domestically-produced white broiler varieties. Although these chickens have not yet become a major factor in Chinese chicken production, they could greatly reduce the country’s reliance on imported breeders in the future (Gu, 2023). Although the implications of a domestic breed of broiler chicken remain to be seen, this development could reduce the cost of Chinese chicken production, which might enable Chinese broiler companies to increase their exports of chicken products. Chinese chicken genetics companies could also emerge as exporters of breeding stock if the domestically bred chickens become popular.

A Growing Industry

Poultry and eggs have only become regular features of meals in China in recent decades (Pi et al., 2014). The most common meat in China is pork, which made up 60% of all the meat consumed in the country in 2021 (McKinsey & Company, 2023). However, meat from white broilers is becoming increasingly popular, a trend the USDA expects to continue (Baych, 2022; Inouye, 2020). Food safety concerns may be contributing to this shift, as could improvements in distribution methods. Many analysts also believe that the government will favor the broiler industry over the pork industry in the future due to more efficient feed conversion ratios (see Pi et al., 2014).

China is home to two of the largest poultry companies in the world, Wen’s Food Group and Wellhope Agri-Tech, which slaughter nearly 1.8 billion birds each year in total (WATT Poultry, n.d.). There are two large-scale producers in China that combine to control about a third of the yellow broiler market, each owning over 100 million chickens (Baych, 2023). It is also possible that other Chinese meat giants will take a more prominent role in the broiler industry. For example, WH Group is already one of the largest pork producers in the world and owns U.S.-based Smithfield Foods. Similar to the broiler industry, the pork industry follows a model of growth based around vertical integration. As it identifies new opportunities for growth and faces increased competition in the pork industry, WH Group has begun expanding its poultry industry (Reuters, 2023; WH Group, 2023; USDA ERS, 1999; Pig Progress, 2021). As broiler meat becomes more common in China, vertically-integrated facilities continue to be built (Baych,

2023). Imports of chickens from Brazil, which already supplies more than 60% of China's imported chickens, are also expected to increase (Baych, 2023).

The Chinese broiler industry is not as concentrated as in the U.S. or Brazil. For instance, in the U.S., most chickens are raised in facilities that produce over 100,000 birds each year, but most Chinese chickens are raised by smaller producers (Pi et al., 2014). In other words, compared to the U.S., China currently has more farms that produce fewer chickens on average. However, these operations are growing in scale. In the late 1990s, the lack of quality agricultural land spurred producers to intensify production further, creating larger and more densely-packed operations. Big producers benefited from economies of scale, leading smaller producers to either go out of business or be absorbed into the production process of the larger companies (Pi et al., 2014). In order to stay in business, companies have scrambled to cut costs as a means of getting a leg up on their competitors. One of the primary ways producers can lower production costs is by skimping on sanitation steps, which has led to food safety concerns.

The government has been an instrumental player in the rise of the broiler industry. In 2006, the Chinese government subsidized nearly $10 million USD in poultry and breeder facility construction, much of which went to large companies (Pi et al., 2014). It has also acted to protect its growing broiler industry. For instance, an outbreak of avian influenza in 2013 was met with almost $100 million in aid from the national government. In addition, state-owned banks provided loans with low interest rates and provincial governments doled out tax breaks and other support (Pi et al., 2014).

The Chinese government has also supported the industrialization of its entire agricultural sector, including the broiler industry, through the designation of "Dragon Head Enterprises" (DHEs), which include corporations like the global grain trading giant COFCO and the aforementioned WH Group. DHEs are large companies that the government has identified as the key drivers of agricultural industrialization. The government provides subsidies and tax breaks to DHEs, and also encourages them to increase their production levels and become more vertically integrated (Schneider, 2016; Pi et al., 2014; Howard, 2019). In many ways, DHEs can be thought of as the Chinese counterparts to Brazil's "national champions" because both sets of companies receive significant government support in order to advance their countries' agricultural standing on the global stage (Béné & Lundy, 2023).

Designated DHEs are tasked with supporting rural development through things like contract growing, which could provide a source of income for rural residents. However, it appears that broiler DHEs are more often contracting with a small number of large-scale grow-out farms (Pi et al., 2014; Howard, 2019). Contract "grow-out" farming has become the standard in the Chinese broiler industry (Xie and Marchant, 2015; Huang et al., 2018). Like in the U.S., integrators provide farmers with chicks, feed, and other production inputs (Xin et al., 2015; Huang et al., 2018).

Production Costs In China's Broiler Industry

As in the U.S. and Brazil, feed is by the far single largest cost of broiler production in China. Feed likely accounts for about 65% of broiler farmers' costs, followed by DOCs, which make up about 18% of costs (Chen et al., 2020). In 2021, the Chinese government's Ministry of Agriculture and Rural Affairs announced guidelines for broiler producers to begin reducing the amount of corn and soybeans used in their feed mixes. This would allow the industry to reduce its reliance on imported feed crops and instead use crops that are more available domestically, such as wheat (Gu & Patton, 2021). More recently, the government has announced a 2025 target of only 13% soybean meal in feed, down from 14.5% in 2022 (Patton, 2023). It's unclear how this will affect the price of feed. Because China is such a massive importer of corn and soybeans, any significant changes in the country's feed crop imports could have big effects on the entire feed industry, which would also affect animal farmers. However, it is difficult to predict what the scale, effectiveness, and repercussions of the government's suggested changes will be (Gu & Patton, 2021; The Fish Site, 2023).

Although some research has suggested that animal welfare is less of a priority in China than in other countries, other research has shown the potential for chicken welfare to become increasingly important to Chinese consumers (Yang, 2021; Sinclair, 2022; Faunalytics, 2022; Faunalytics, 2022). If concern for animal welfare grows, producers may upgrade their facilities or take other steps that would increase their production costs.

Currently, demand for broiler chicken meat is mostly concentrated on the country's eastern coast. China's distribution network generally struggles to keep chicken products cold over long journeys, so the broiler industry is largely based in close proximity to its customers on the coast. If improvements were made to the distribution network, the industry could begin moving to Central China where wages are lower, which would lower production costs (Pi et al., 2014).

The Chinese Egg Industry

China is the world's largest producer of eggs and relies very little on imports from other countries to meet domestic demand (FAO, 2023a). The country is home to one of the five largest egg producers worldwide, Beijing Deqingyuan Agricultural Technology Co. Ltd., which owns over 20 million layer hens. Other major Chinese egg producers include Sichuan Sundaily Farm Ecological Food Co. and Shanxi Jinlong Breeding Co., which own 13 million and 12 million layer hens, respectively (Clements, 2022).

Egg production is mostly concentrated in North and Northeast China, especially in Hebei, Henan, Liaoning, and Shandong provinces, which combine to produce over 1.5 million metric tons of eggs (Yang et. al, 2018; Yang, 2021;,Windhorst, 2018). These provinces, along with Sichuan Province and Jiangsu Province, produce between 50% and 60% of the country's eggs (Yang et. al, 2018; Yang, 2021). The vast majority of eggs in China are brown or light brown, with only a tiny fraction being white or blue eggs (Yang, 2021; Yang et. al, 2018; Chen et al., 2023). The U.S.-based multinational Hy-Line supplies about three-quarters of the layer hen

stock in China, with domestic companies Jinghong and Jingfen supplying another 12% (Yang et. al, 2018).

Similar to the broiler industry, the Chinese egg industry has grown in stages, largely due to shifts in government policy. After the 1978 Reform and Opening-Up policies were implemented, a period of fast development took place (Yang, 2021; Chen et al., 2023). From 1991 to 2000, egg production skyrocketed. Since then, egg production has generally increased at a slower rate (Yang, 2021).

Today, the Chinese egg industry is composed of a mix of small- and medium-scale producers as well as large, vertically integrated producers who resemble U.S. producers in their control of everything from feed milling to egg distribution. Although small producers still contribute significantly to the country's total egg production, comparatively higher costs, shrinking opportunities in rural areas, and government support for intensification are combining to drive these small farmers out of existence (Yang et. al, 2018).

Egg Production

Egg production methods can vary greatly between smaller and larger egg producers. Many large egg companies have invested in automated systems for feeding and collection, while smaller farmers still rely almost exclusively on less efficient manual feeding and collection. For instance, 85% of producers with over 100,000 layer hens use automated feeding systems. Manual feeding is used by 90% or more of producers with under 9,000 layer hens (Yang et. al, 2018). Large farms use hens for laying periods ranging from 60 to 70 weeks, but smaller producers are less standardized (Yang et. al, 2018). As the industry consolidates and standardizes, the more efficient approaches used on larger farms will likely become the norm.

Unlike in many other countries, egg farmers in China produce table eggs almost exclusively. In the U.S. and around the world, about 30% of eggs are processed for use in products like baked goods rather than being sold as table eggs. In China, this is true for less than 5% of egg production (Yang, 2021). Rather than the cartons containing a dozen eggs that are familiar to U.S. consumers, eggs in China have typically been sold without packaging and name brands, though these may increase the profits of egg producers (Yang, 2021; Yang et. al, 2018). Companies are increasingly cleaning, grading, and packaging eggs in branded containers (Yang, 2021).

About 90% of eggs are produced using cage systems, with most of the remainder coming from free range systems (Yang et. al, 2018; Chen et al., 2023). Over 85% of egg farms use a tiered stair cage system, which reduces the costs involved in labor, space, and sanitation. However, vertical cage systems that allow for a higher density of chickens and can be outfitted with automated waste collection may grow in popularity (Yang, 2021).

The manure from vertical cage systems may be more easily used for fertilizer and biogas, two ways of reusing waste that the Chinese government has shown interest in. In fact, all farms with over 100,000 chickens, currently comprising less than a percent of poultry farms, are required to

install digesters to produce biogas (Barclay, 2010). Farms are also required to avoid the production of excessive odor from manure. This manure can be processed into fertilizer, which farmers may sell for use in fruit and vegetable farming (Yang, 2021).

The Chinese government has also encouraged research on genetics to develop domestic breeds that can lay more eggs or have more efficient feed conversion ratios, all designed to lower costs for producers. Today, more than half of layer hens are domestic breeds (Yang, 2021).

Consumers And The Retail Market

Within China, eggs are the second most common form of animal protein, trailing only pork (Yang, 2021). One reason eggs are so popular is because they are significantly cheaper sources of protein than other animal products. However, even if prices increased, eggs would likely remain common among the Chinese public. Demand for eggs in China is fairly inelastic, especially among wealthier segments of the population, so higher prices don't greatly affect the number of eggs sold (Yang, 2021; Yang et. al, 2018).

Eggs are mostly purchased in urban areas, but rural residents are increasing their consumption (Yang et. al, 2018). Shoppers are increasingly able to purchase eggs from supermarkets, but the most common retail source of eggs are small shops that specialize in poultry products (Yang et. al, 2018). E-commerce is also a common way for Chinese consumers to purchase eggs (Yang, 2021).

Familiarity With Cage-Free Eggs And Animal Welfare

A 2021 study of eggs in major supermarkets across four of the largest cities in China found that 17% of eggs claimed to be either cage-free, free range, or organic (Chen et al., 2023). Due to the possibility of false advertising, it is difficult for consumers to know if labels on eggs, when they are packaged at all, are accurate (Chen et al., 2023). Moreover, the term "cage-free" does not appear to be commonly used in China and one study found that a third of the Chinese public has never heard the term "cage-free" (Chen at al., 2023). Free range eggs are more commonly advertised and some groups, including people with higher incomes, may be willing to pay more for them.

On a broader level, the phrase "animal welfare" (*dòngwù fúlì* 动物福利) has rapidly become more familiar to the Chinese public. In 2011, two-thirds of people said they had not heard of "animal welfare," a number that dropped to only about 25% in 2019 (Chen et al., 2023). However, despite that greater familiarity, consumers' understanding of what "animal welfare" means for farming standards still appears low (Faunalytics, 2022). In recent years, industry-influenced guidelines for layer hen welfare both in and out of cages have been developed, as have awards for hen welfare (Chen et al., 2023).

Chinese producers have noted that they are unsure if consumers would be willing to purchase costlier eggs. Other barriers include "the absence of trustworthy food labeling, lack of land,

restrictions related to environmental policies, and challenges to managing cage-free systems" (Chen et al., 2023). It is also unlikely that international markets would influence layer hen welfare in China given the country's minimal import and export of eggs (Yang et. al, 2018; FAO, 2023a; Chen et al., 2023). All of this suggests that cage-free eggs could eventually become more popular in China, though significant barriers still exist.

The Future Of The Chinese Egg Industry

The Chinese egg industry is forecasted to grow increasingly concentrated, standardized, and industrialized, largely driven by the goals of the government. However, with a sizable proportion of the country's egg supply still coming from small or medium producers, scaling will not be cheap. Building larger, more automated facilities requires significant capital investments, which could be partially paid for by the government or foreign companies (Windhorst, 2018). Like other industries, feed costs, feed conversion ratios, diseases like HPAI, waste treatment, pollution, and food safety will remain important factors in the cost of production (Windhorst, 2018; Yang, 2021). Like broilers, corn and soybeans continue to be the dominant feed ingredients for layer hens (Yang, 2021). However, as noted above, this could change in the future as a result of government policies aimed at reducing the prominence of corn and soybeans in feed.

The Chinese Aquaculture Industry

Although China has a long history with aquaculture, its modern aquaculture industry did not begin in earnest until the 1980s (Crona et al., 2020; Chen & Gao, 2023; Farquhar et al., 2017). Today, China is the world leader in total fish production and shows no signs of letting up. The country has been the top aquaculture producer since 1991 and produced 57% of the world's aquaculture-raised animal products in 2020 (FAO, 2022). The global seafood trade, including aquaculture, is now larger than the chicken and pork industries combined, further underscoring the central role China plays in the animal agriculture industry (Little et al., 2018).

In terms of fishes produced by the Chinese aquaculture industry, carp are far and away the most common. In 2021, the industry produced 19.1 million metric tons of carp, mostly for domestic consumption. Carp aquaculture occurs in most provinces in China and the animals are typically raised in ponds, lakes, or reservoirs (FAS China Staff, 2023). The next most common fishes are tilapia (1.7 million metric tons), which is largely produced for export, and catfish (1.3 million metric tons), a growing subindustry (FAO, 2023b; Crona et al., 2020; FAS China Staff, 2023).

Production Costs

Costs can vary widely because production systems in the Chinese aquaculture system are quite diverse, ranging from lake and reservoir farming to sea cages or pens (Cao et al., 2015; Engle et al., 2019). Earthen ponds, which are comparably cheap to operate, are the most common production method (Godfrey, 2019; Little et al., 2018). However, pond production and similar methods can have considerable environmental impacts and are more difficult to scale up. They also require large amounts of land, water, and other limited resources.

Given government policies that encourage large-scale production, the future of the Chinese aquaculture industry will likely be one of increasing intensification of production methods (Godfrey, 2019). Unlike traditional production, intensive fish farming requires the use of specialized feeds, as well as water exchange and aeration in order to keep the water hospitable for fishes. As long as there is sufficient demand for aquaculture products, these additional costs, plus the high cost of capital like tanks and machinery, will likely be offset by the higher production resulting from increased fish densities and industry concentration (Cao et al., 2015). Breeding and other genetic manipulation may also decrease costs, especially for fishes like tilapia, which reproduce fairly quickly (Little et al., 2018).

In addition to terrestrial crops like soybeans and corn, a costly ingredient for fish feed is fishmeal. Only 3% of carp feed and 6% of tilapia feed comes from fishmeal, but because Chinese carp and tilapia farming is so widespread, the cost for this ingredient can add up quickly (Cao et al., 2015). Fishmeal is the most expensive ingredient in many types of fish feed and usually comes from wild caught fishes. These fishes are often types that humans don't eat, but they can also be juveniles from species that humans do eat. Because of this, and likely to reduce costs as well, the Chinese government has started promoting "compound feeds," which include more plant ingredients and less fish-based ingredients than traditional feed (Chun, 2022; Ministry of Agriculture and Rural Affairs, PRC, 2021; Ministry of Agriculture and Rural Affairs, PRC, 2022).

Chinese aquaculturists are also likely working to improve feed conversion ratios for carp and tilapia, both in terms of genetics and improved formulas for feed (Little et al., 2018). Currently, in China carp and tilapia have feed conversion ratios of 1.7 and 1.6, respectively, meaning it takes 1.7 kg of feed for a carp to produce 1 kg of flesh used for meat (Cao et al., 2015). By contrast, the Norwegian Atlantic salmon industry uses just over 1 kg of feed to produce 1 kg of meat (Asche et al., 2018). More efficient feed conversion ratios save farmers considerable amounts of money because they do not have to purchase as much feed to raise large fishes.

Industrialized Aquaculture in China

Highly industrialized production is poised to become a much more central part of China's aquaculture industry in the coming years. More traditional approaches such as pond farming can cause environmental pollution, food safety issues, and a variety of other concerns (Chen & Gao, 2023). The Chinese government has imposed a number of sustainability regulations that make traditional approaches more difficult (Newton et al., 2021; Godfrey, 2019). If the government's approach to the rest of the meat industry is any indication, future policies will also encourage intensification and vertical integration (Pi et al., 2014). Increased production costs are also driving many producers to become more efficient, which often involves intensifying their methods (Godfrey, 2019; Chen & Gao, 2023; Newton et al., 2021). As the industry continues to advance, approaches like RAS will likely grow more common.

In 2021, freshwater aquaculture accounted for 57% of the country's industrial aquaculture farm area use, with the remaining 43% used for seawater production (Chen & Gao, 2023). These

modes of production contribute fairly equally to China's overall industrial aquaculture production total. Much of this production comes from just two provinces: Shandong and Fujian, which contribute 37% and 20% of China's total industrial aquaculture production, respectively (Chen & Gao, 2023). Freshwater aquaculture operations mostly take place in Fujian, Shandong, Hubei, Jiangxi, and Anhui provinces, while seawater operations are largely concentrated in Shandong, Liaoning, Fujian, Jiangsu, and Hainan provinces. Overall, aquaculture production nearly tripled between 2003 and 2021 (Chen & Gao, 2023).

At present, the more industrial segment of Chinese aquaculture focuses on fishes such as turbot and half-smooth tongue sole, which are especially common in industrialized aquaculture, as well as fishes and invertebrate species including grouper, large yellow croaker, giant river prawns, white shrimps, and starry flounder. Other aquatic animals commonly farmed in industrial settings include mud crabs, eels, rainbow trout, abalone, and sturgeon (Chen & Gao, 2023).

There are three main production types in Chinese industrial aquaculture: flowing water systems, RAS, and aquaponics systems. Flowing water systems are common in turbot and sturgeon production and typically use underground seawater to fill fish tanks equipped with aerators (Cang et al., 2018; Chen & Gao, 2023; Shen et al., 2014; World Organisation for Animal Health, 2009). This type of production is resource-intensive, can be difficult to standardize, and results in the discharge of dirty wastewater directly into the surrounding environment (Chen & Gao, 2023).

As discussed in the U.S. aquaculture section, highly productive RAS tanks involve the reuse of filtered water, which can save producers money in the long run though upfront costs to build them can be high. Although they are becoming more prevalent, these recirculating aquaculture systems still only contribute to a small fraction of China's total aquaculture production (Chen & Gao, 2023).

Producers may also explore aquaponics as a greener, more cost-effective approach to aquaculture. Aquaponics systems attempt to minimize the environmental impacts of aquaculture using vegetable farming to absorb waste (Chen & Gao, 2023; Cohen et al., 2018).

Barriers To Increasing Intensification

Because the global aquaculture industry is still fairly young compared to many other animal agriculture industries, there are many ways its costs could change. Currently, there are several barriers to continued expansion of industrial aquaculture in China, including production costs, sale prices, technological limitations, and labor issues. As is true in other countries, Chinese aquaculture must grapple with feed costs, high capital costs for construction and machinery, and the high cost of energy needed to maintain conditions and treat waste.

Currently, fishes raised in particularly intensive systems such as RAS production do not consistently sell for enough money to cover the high cost of production, especially when compared to less intensive forms of aquaculture like pond production. Even if more intensive systems like RAS were built, there would be a shortage of workers with the specialized

knowledge required to run these operations (Chen & Gao, 2023). Technological innovations and government policies could lead to lower costs, more qualified laborers, and the increased use of RAS, but currently over 95% of China's industrial aquaculture farm area uses less advanced techniques involving flowing water production (Chen & Gao, 2023; Fletcher, 2021; Ying et al., 2015; Song et al., 2019). However, RAS is one of the aquaculture approaches that most clearly resembles land-based factory farms, so they are an important feature of the industry for advocates to be familiar with.

Despite the current profitability struggles, industrialization and intensification in aquaculture may become more profitable as the industry matures. Changing consumer preferences around health and environmental impact may spur a shift toward systems like RAS and aquaponics. Direct government support—like with Dragon Head Enterprises in the broiler industry—is always a possibility in China, as is foreign investment. Either funding source would make it more feasible for companies to construct intensive operations or even expand the geographic scope of the industry.

The Future Of A Developing Industry

Technological innovations can similarly improve profitability and could be driven by government-sponsored research groups like the Chinese Academy of Fishery Sciences (Farquhar et al., 2017). As the industry and employment opportunities grow in China and abroad, it is likely that training programs for managers and operators will become more common.

The central government's Five-Year Plans—economic development plans that lay out China's priorities for the coming years—can also provide insight into the probable direction of aquaculture in the future. For example, the current 14th Five-Year Plan emphasizes goals like making aquaculture more environmentally friendly and the integration and modernization of the fishing industry (Ministry of Agriculture and Rural Affairs, PRC, 2022; Aquaculture Magazine, 2022).

In 2020, new government standards limiting the use of veterinary drugs like antibiotics in aquaculture also went into effect, which could increase costs for producers (FAS China Staff, 2023). This is reinforced by demand for strict food safety standards in countries that consume Chinese exports (Little et al., 2018). Producers in China may move to follow producers in the Atlantic salmon industry, who have largely phased out antibiotics in favor of preventative vaccines (Asche et al., 2018).

In following the path laid out by more advanced industries like that of broiler chickens, Chinese aquaculturists may look to increase the number of value-added products—further processed items like pre-seasoned or breaded chicken—or other products they can create from farmed fishes (Asche et al., 2018; Backas, 2003). While the chicken industry has products like nuggets and tenders that increase sales and reduce waste, thereby decreasing costs, the aquaculture industry is not as diversified, especially for markets outside of Asia (Asche et al., 2018).

Creating new products that consumers will purchase, as well as developing the specialized machinery needed to make those products, could lead to additional growth in the industry.

Even in the fairly advanced Norwegian Atlantic salmon industry, line speeds at large processing plants reach about 6,500 fishes per hour, far slower than the U.S. broiler industry, which is capped by law at between 8,400 and 10,500 chickens per hour (Asche et al., 2018; Union of Concerned Scientists, 2019). The aquaculture industry may work to improve these speeds as well, which would also lower costs. Industry experts note that breeding efforts to create fishes with more consistent sizes and appearance could both reduce costs and improve sales (Asche et al., 2018).

The aquaculture industry could grow even faster if it can replicate the production methods and body of research the broiler industry already has at its disposal. For instance, capital-intensive methods like automated machines can lead to massive productivity increases compared to more labor-intensive methods, as evidenced by the U.S. broiler industry (Asche et al., 2018). Chicken products are also faster to process and take longer to spoil than fish products, which the aquaculture industry may address with future research and development. In addition, agricultural research on chickens is abundant and still growing. Chicken producers have techniques to change behavior using light, affect feed consumption using temperature and humidity controls, and so on. Because the farmed fish industry is younger and focused on a wider variety of animals, this information is less readily available to aquaculturists. Even when available, it can be expensive to implement this knowledge (Asche et al., 2018). Fish products also often take longer to process and spoil more quickly than chicken products, which future industry research and development may seek to address.

In general, reducing costs will likely lead to lower prices, which may spark increased demand and additional production. In other words, if the Chinese aquaculture industry is able to lower costs, it could soon be exploiting fishes in even greater numbers (Asche et al., 2018). Vertical integration and consolidation, which are already underway, will increase the speed of this process.

Conclusions

Animal Agriculture Is An International Industry

The economy is global. This is especially true of most aspects of animal agriculture. Rather than small family farms feeding local communities across the world, the reality is that a small number of multinational megacorporations now dominate nearly every aspect of the industry. U.S. corn is fed to farmed animals in Mexico. Brazilian meat processors own giant U.S. chicken companies. German animal genetics companies supply Chinese corporations modeled after those in the U.S.

If animal advocates want to be strategic and effective, it is critical to have at least a foundational understanding of this complex web of money and animal suffering. This is not to say that domestic campaigns are no longer important aspects of the movement. Instead, it is a call to consider how actions in one country can shape the industry in another country. Advocates living in major producer countries like the U.S., Brazil, and China, or those living in regions with large amounts of purchasing power, like the E.U., can and should seek to change industry practices around the globe.

The animal agriculture industry will need to make changes to its production practices as consumers grow more concerned about animal welfare. It's likely that every reform aimed at improving conditions in animal agriculture would cost the industry more money, such as transitioning to cage-free housing for layer hens or requiring stunning for fish slaughter. As a result, advocates campaigning for welfare reforms can simultaneously increase costs for the meat industry while also marginally reducing the suffering of farmed animals.

In the globalized animal agriculture industry, producers in one country often have buyers in another. As a result, consumers in one part of the world can demand certain practices of their producers and, unless they are able to find another market, the producers will be obligated to meet these demands (European Commission, 2018; European Parliament, 2022). For example, consumers in the E.U. have demanded—and received—certain animal welfare standards for fishes coming from Vietnam (Little et al., 2018; Lines & Spence, 2014). Similarly, regulatory agencies that oversee imports could require that animal products must come from animals that are raised without antibiotics.

Consolidation And Vertical Integration Are Hallmarks Of The Industry

A small number of animal agriculture megacorporations control much of the industry. For example, about one out of every six chickens slaughtered in the U.S. comes from Pilgrim's Pride. But in this industry, even a corporation as big as Pilgrim's is owned by an even bigger corporation: Brazilian-owned JBS, the largest meat producer in the world. Similarly, recognizable U.S. brand names like Ball Park, Jimmy Dean, and Hillshire Farm are all owned by Tyson, which gives consumers the illusion of choice in a more limited market.

Besides buying other meat-producing companies, one of the main ways that Tyson, JBS, and other giants were able to reach their current "megacorporation" status is through vertical integration. By not only controlling other meat processors, but also controlling the suppliers of feed and day-old chicks, as well as everything from distribution networks to marketing teams, these companies are able to be much more efficient with their operations than their competitors. This ultimately saves them considerable amounts of money and solidifies their standing atop the industry.

Although governments sometimes move to block acquisitions that would further increase the power of these companies, they often do little to limit their growth or break up their strangleholds over markets. Strengthened antitrust regulations, which are intended to promote competition in markets by preventing things like monopolies, are a potential tool for governments to use to curb the power of these megacorporations. Advocates could lobby government officials to shift or strengthen current policies in order to limit the industry's reach. More often, however, governments around the world provide both direct and indirect support to the animal agriculture industry. In some countries, like Brazil and China, governments have even actively facilitated the consolidation of the industry.

It is important to note that different sections of government play different roles in this dynamic. For example, in the U.S., the Department of Justice and the Securities and Exchange Commission are two of the main bodies tasked with overseeing the legality of business dealings, while the USDA is one of the main agencies involved in promoting the animal agriculture industry, though it also has a regulatory role. Advocates may need to focus their efforts on specific agencies depending on the change they are demanding. For instance, petitioning the USDA for limits on corporate mergers would have little effect because the Department of Justice oversees this type of action. Because there are a number of regulatory bodies overseeing different parts of the industry, advocates have several routes available to them and can investigate which avenues may be the most fruitful.

Will Aquaculture Become The Chicken Of The Sea?

Of all the industries covered in this report, the U.S. broiler industry provides the most extreme example of intensification. Over nearly a century, it has grown into a highly consolidated industry dominated by just a small number of fully vertically integrated corporations. Through careful breeding and management of genetics, chickens used by these companies now grow faster and larger, better endure horrific farming conditions, and cost less to raise. Moreover, the industry is standardized to a great degree, with automated equipment churning out billions of chicken-derived products each year. In addition to standard pieces of chicken meat like breasts and thighs, the industry has pioneered many value-added products. For example, there are chicken tenders and nuggets, which are made with specialized equipment and use more parts of the chicken, thereby decreasing waste and costs while making the broiler processors more money. There is precedent for other animal agriculture industries trying to copy these money-making aspects of broiler chicken production as well: many other industries, especially pork, have sought to replicate them.

By contrast, the modern aquaculture industry is quite young and unstandardized, which means it lacks many of the cost-saving efficiencies of the broiler industry. It uses a wide range of fishes and other aquatic animals, many of which are still raised in earthen ponds. Though many large companies are involved, they are often significantly smaller than the companies dominating the U.S. and global broiler chicken industries. Specialized equipment and agricultural knowledge exist in the aquaculture industry, but are rarely as effective or extensive as in industries that

exploit chickens. Few value-added products come out of the aquaculture industry, and more standard products are also less consistent in size and other characteristics compared to the broiler and egg industries.

However, all of this is changing. As the industry consolidates further, grows increasingly vertically integrated, and breeds more “efficient” fishes, the aquaculture industry as a whole will likely grow to resemble the broiler industry more and more. Larger companies will spend more money on capital investments, such as land-based recirculating tanks, allowing them to reap the rewards of economies of scale. Research and development will improve feed conversion ratios and create additional automated machinery. New value-added products will be introduced.

This replication isn’t inevitable. Advocates can see where the aquaculture industry is headed. As a result, they have a unique opportunity to affect its trajectory and prevent the expansion of an already massive system based on animal suffering.

Costlier Animal Products Require More Accessible Alternatives

There are many opportunities for advocates to increase the costs of animal products. This would both limit the profitability of animal exploitation and, by raising prices, make animal products less appealing to consumers. However, advocates taking this approach must also be aware of the need to increase access to plant-based alternatives. Many animal products have fairly inelastic demand, meaning that price increases won’t greatly decrease the amount of that product people buy. This is not because the price isn’t a concern, but rather because these products are staples for many people: they see them as irreplaceable. Often the first people who are priced out of these products are those who are already subsisting on low incomes, so advocates need to ensure that increased prices for animal products are met with decreased prices for plant-based products (Kenton, 2023). Without improved access to plant-based products of all types, creating costlier animal products would compound the harms marginalized people are already subjected to.

Successful efforts to improve welfare standards are one way to increase animal agriculture companies’ production costs. These changes would also likely result in higher retail prices. Although higher prices alone may not be enough to shift consumer preferences toward plant-based options, it is important that advocates work to make plant-based products more affordable and widely available (Rethink Priorities, 2023). Improved accessibility is a primary requirement for a plant-based future and must happen at the same time as animal products become less financially appealing to the general public.

Aquaculture Is A Collection Of Industries

The broiler and egg industries use only a handful of varieties of chickens, most of which come from a small number of chicken genetics companies. Aquaculture, by contrast, involves hundreds of types of fishes and other aquatic animals, as well as dozens of different rearing

techniques. Aquaculture includes everything from salmon to tilapia, sea cages to recirculating tanks, freshwater to saltwater. Although monocultures (i.e., the production of only one species in a system) are becoming increasingly popular, there is a huge amount of diversity in the industry, as well as in consumer preferences.

One possibility is that the industry will further consolidate around the most commonly farmed fish species: salmon, tilapia, carp, and catfish. However, even those fishes are considerably more different from one another than the different breeds of broiler chicken in use. It may be useful for animal advocates to begin thinking less about the aquaculture industry and more about the salmon industry, the tilapia industry, and so on. Advocates can learn more about these industries through the research conducted by groups like the Fish Welfare Initiative.

Advocates Lack Resources For Understanding The Animal Agriculture Industry

Much of the animal advocacy movement's research that has focused on the animal agriculture industry has looked at its animal welfare practices. This work is unquestionably crucial. But to supplement this body of knowledge, the movement needs more advocate-led explorations of other aspects of the animal agriculture industry, like the complexities of the production process. Regardless of advocates' ultimate goals as they relate to the industry, understanding how it operates is necessary in order to change it.

This lack of information is especially noticeable as it relates to large producer countries other than the United States. While the scope of this research was limited to English—as noted in the Caveats & Limitations section below—we suspect that few resources exist in languages like Portuguese and Chinese as well. The animal advocacy movement can greatly benefit from sharing research and resources across languages, especially because of the global nature of the animal agriculture industry. For instance, advocates working in Brazil need access to information about what is happening in the U.S. and in China.

Advocates need to know more about the ins and outs of these industries, as well as the interdependent feed industry, but this is difficult when there is such little information compiled for advocates by fellow advocates. Funders and multilingual advocates can support the movement by translating resources or publishing an overview of the information available in other languages.

Detailed information on aquaculture is particularly lacking, likely because the industry is so new and diverse. In order for advocates to get out in front of the consolidation, standardization, and expansion of aquaculture, more resources exploring the practices of the industry are necessary. In other words, it's important for advocates to expand their knowledge of aquaculture before it's too late.

Opportunities For Advocate Intervention And Collaboration Abound

As this report aims to show, the animal agriculture industry has potential points of intervention at every stage of the supply chain. From working to reallocate crop subsidies to improving slaughter conditions, advocates can take steps to reduce and even reverse the industry's ability to exploit animals. An understanding of the industry's economics reveals even more pressure points. However, because there are so many places to intervene, advocates would benefit from further analysis of which of these points are most promising.

There are also countless opportunities for collaboration across movements. For instance, animal advocates' goals share significant overlap with those of groups focused on climate change and the environment: not just greenhouse gas emissions from animal agriculture, but also several other less-often discussed issues such as single-use plastics and environmentally-unfriendly packaging, as well as pollution from feed crops or the global transportation of animal products. To this end, forthcoming Faunalytics research will highlight opportunities for collaborations between the animal protection movement and the climate and environmental movement.

Because workers are often exploited in the animal agriculture industry and are a main cost for corporations, there may similarly be opportunities to work with labor groups such as unions. Grow-out farmers, especially those who would like to exit the industry but are too indebted, may be another group of laborers to seek out as partners in working to fight against the big animal agriculture companies. Some people who have historically been viewed as advocates' enemies may be trapped in the system by debt or their limited alternatives for making a living.

Feed and Genetics Are At The Center Of These Industries

Feed is usually the main factor affecting the cost of production of animal products, including chicken meat, eggs, and fish products. This often indicates that the industries are highly advanced and have become very efficient in other aspects of their operations, aided by innovations like automated machinery to limit labor costs. A huge share of the world's supply of two crops, corn and soybeans, is used to feed confined animals, and much of these crops' global trade takes place among the U.S., Brazil, and China. Especially in the U.S., government policies make these inputs cheaper for the animal agriculture industry, thereby supporting animal suffering.

Because feed is so costly for the industry, companies have focused on improving feed conversion ratios so that animals grow larger or produce more eggs with less feed. In addition to specially formulated feed blends, advances in feed conversion efficiency has come from genetic management, including with selective breeding. Other focuses of selective breeding and the genetics companies that carry it out include faster growth and tolerance for poor conditions. In both the highly efficient chicken industries and the growing aquaculture industry, genetic

advantages such as these will likely remain at the forefront of efforts to become increasingly profitable. Increased efficiency often often means increased suffering for animals. Limiting genetic manipulation and cutting subsidies for animal feed crops could both help prevent the expansion of the industry.

Room For More Research

There are many directions for future research to take when considering the factors affecting the cost of production for chicken, fish, and egg products. The following are some of the directions we see as most useful, but it is not a definitive list.

One option for research is to map out the production process in other high-production countries. The U.S., Brazil, and China are three of the largest players in global animal agriculture, but far from the only ones. Other major producers of chickens, eggs, and fishes include countries like India, Indonesia, and Russia. Because supply and demand are inextricably linked, research could also focus on the biggest consumers of animal products even if they aren't major producers.

In general, there seems to be a divide in terms of both economic and industrial development between the large producer countries and countries that produce fewer animal products. In the U.S., Brazil, and China, many of their animal agriculture industries are highly sophisticated and dominated by megacorporations. Other countries that produce massive quantities of animal products are often similarly industrialized. In countries with fewer large-scale producers, the production process is likely quite different and the amount of animal suffering may be much less as well. However, like the broiler industry in Brazil or the aquaculture industry in China, rapid animal agriculture expansion and intensification could occur in any of these countries. Understanding the local context could enable advocates to prevent the industry's growth in these countries.

Advocates would also benefit from deeper individual analyses of the industries we cover in this report. This report provides an introduction to many aspects of the chicken, fish, and egg industries around the world, but there is still a lot of information we did not cover about their specific processes and the regulations that govern them. Such resources would provide advocates with a more thorough map of the animal agriculture industry, revealing new points of intervention in the process.

Future research could also map other industries, such as the pork, beef, and dairy industries. In many ways, these industries resemble those covered in this report. For example, many of the practices used in the modern pork industry were directly modeled off of the broiler industry. China, the U.S., and Brazil are also the three leading slaughterers of cows and all three are among the five largest slaughterers of pigs. However, each industry has its own unique factors that are important for advocates to understand. Moreover, Pakistan is one of the largest slaughterers of cows and is especially under-discussed in research on this topic. Additionally, the E.U. is an important player in the global meat trade, and is home to two of the largest

pork-producing countries, Spain and Germany. As a result, explorations of other industries would also provide important background for advocates operating in parts of the world other than North America, South America, and East Asia.

Finally, researchers could also investigate countries that support animal agriculture by importing products rather than producing them. Global animal agriculture couldn't exist without customers who buy its products. As a result, future research should examine the countries that import the most animal products. For example, Japan is among the top importers of chicken and fish products, while Germany is one of the biggest importers of chicken, fish, and egg products. With a stronger understanding of the context in countries like these, advocates can work to shift consumer preferences to prioritize animal welfare. Advocates in these countries can also call for regulations that require better animal welfare practices in the production of imported products. If customers in these countries demand better treatment of animals, their suppliers in other countries must comply or else risk losing business.

Caveats & Limitations

As with all reports, this one has some important caveats and limitations to bear in mind.

Many of the practices of the animal agriculture industry are trade secrets and are not available to the general public. As a result, there may be industry processes or ongoing research and development that are not covered in this report. Similarly, most of the cost estimates in this report come from academic studies and estimates rather than directly from the industry itself.

Although some information is limited, many of the factors affecting the cost of production of animal products are described in painstaking detail in other resources. Entire books have been written about some of the industry practices topics covered in this report. In other words, this report tries to summarize as many of the industry's considerations as possible, particularly the ones that could be actionable for animal advocates. However, there is still a significant amount of information not included here. Similarly, the economic mechanisms of markets for animal products have been simplified to emphasize key effects.

The scope of this report was restricted almost entirely to English-language resources due to our own staff limitations. As a result, there is likely additional information that could be included in future reports—particularly for Brazil and China—but which we were unable to find and include.